

A1: RESEARCH QUESTION

What features affect our customers and cause them to Churn?

A2: GOALS

The main goal of the data analysis is to determine what features are important in influencing whether a customer in a telecommunication industry will churn.

According to Bhandari, Pritha (2023, June 22), the independent variable is the cause. In contrast, the dependent variable is the effect, and its value depends on the changes of the independent variable in a cause-effect relationship.

Using the Logistic regression analysis, this analysis aims to determine which features significantly impact the churn.

B1: SUMMARY OF ASSUMPTIONS

Logistic regression is a method that is used to fit a regression model when the response variables are binary and are used when the response variable is logical. According to Bobbitt, Zach (2020, October 13), a regression model has the following assumptions.

1. The response variable in a logistics regression only has two possible outcomes hence it is binary
2. The observations are independent of each other
3. There is no high correlation or multi-collinearity between the explanatory variables.
4. There are no extreme outliers in the observations.
5. The sample size is sufficiently large enough to draw valid conclusions from the fitted model.

B2: TOOL BENEFITS

I will utilize the Python tool to undertake the logistic regression analysis. Python is an open-source general-purpose programming language with many packages that can be used and shared with others. I utilized Python Programming Language because of its flexibility and extensive library collections, which are valuable for performing complex calculations and Logistic regression analysis. The libraries will be used to import data into the dataframe, observe the data in the dataframe, perform statistics, convert categorical data to numerical data, create the visualization, and perform logistics regression analysis. The following libraries and packages will be utilized

1. Data Cleaning and Exploration

- Numpy
- Pandas
- Statistics
- Scipy

2. Visualization

- Seaborn
- Matplotlib
- Pylab
- Scikitplot

3. Regression Analysis

- Sklearn.model_selection (for train-test split)
- Sklearn.linear_model (for logistic regression)
- Sklearn.preprocessing (for StandardScaler, LabelEncoder, OneHotEncoder)
- Sklearn.metrics (for accuracy, confusion matrix, ROC metrics)

- Sklearn.feature_selection (for RFE and SelectFromModel)
- Statsmodels (for statistical modeling and variance inflation factor)

4. Handling Warnings

- Warnings

5. SMOTE and Model Feature Selection

- Imbalanced-learn (for SMOTE)
- Sklearn.feature_selection (for SelectFromModel)
- Math

B3: APPROPRIATE TECHNIQUE

According to Thanda, Anamika(2023, May 11), Regression analysis is a type of predictive modeling technique that is used to find the relationship between a dependent variable (usually known as the “Y” variable) and either one independent variable (the “X” variable) or a series of independent variables. When two or more independent variables are used to predict or explain the outcome of the dependent variable, this is known as multiple regression. Logistic regression is an analysis used to calculate (or predict) the probability of a binary (yes/no) event occurring. Logistic regression is a classification algorithm that predicts a binary outcome based on independent variables.

Since the analysis aims to determine the significant features that impact the Churn, a categorical variable with binary values (yes/no), logistics regression analysis is the ideal technique. More so, to be able to analyze all the variables, the categorical data will be converted to numerical data since logistics regression analysis analyses the relationship between quantitative or numerical data

C1: DATA CLEANING GOALS

After loading the data, it was viewed and reviewed to understand its nature.

1. Data type constraint and Data Missingness

Due to the wrong datatype specified while loading the Internet service, the value of Other was identified as missing. To address the issue, the data type was specified as missing data. To address the issue, the value feature was specified as a String

2. Data Uniqueness

The data was assessed to determine if there were any duplicate values. No duplicate values were found in both the columns and the rows.

3. Drop redundant features with high cardinality and features with high collinearity. The features of CaseOrder, Customer_id, Interaction, UID, City, State, County, Zip, Lat, Lng, Job, TimeZone, Churn, Bandwidth_GB_Year, Tenure.

4. Check for Outliers. I will find Outliers in the data set by plotting to identify outliers or outlying observations. Due to a lack of understanding of the domain of the data provided/other pertinent information regarding the data, the features with outliers were retained.

5. Convert Categorical data to numerical data through dummy encoding (One Hot encoding)

According to Saxena and Shipra (2024, September 18), preprocessing categorical data is necessary since most machine learning models only accept numerical data. One way to do this is through one hot encoding, which is used when the features are nominal, and each category is mapped with a binary variable 0 or 1. 0 represents absence, and 1 represents the presence of that category. The newly created binary features are known as Dummy variables. Dummy encoding uses N-1 features to represent N labels/categories.

C2: SUMMARY STATISTICS

According to Look, Marie (2024, April 1), A dependent variable, often represented as Y, is the variable that is observed and measured to determine the experiment's outcome. An independent variable, often denoted as X, is the variable that is intentionally manipulated or controlled by the researcher. It's the factor that researchers believe may have a causal effect on the dependent variable.

1. Churn

The data type for Churn is categorical. The Churn variable will be used as a dependent variable

```
print(Churn_df[['Churn']].describe())  
print(Churn_df['Churn'].unique())
```

```
      Churn  
count  10000  
unique      2  
top      No  
freq    7350  
['No' 'Yes']
```

2. MonthlyCharge

The data type for the monthly charge feature is numerical. The monthly Charge will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      172.624816
std       42.943094
min       79.978860
25%      139.979239
50%      167.484700
75%      200.734725
max       290.160419
Name: MonthlyCharge, dtype: float64
```

3. Area

The data type for the Area variable is categorical. The area variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Urban' 'Suburban' 'Rural']
Summary statistics:
count    10000
unique      3
top      Suburban
freq      3346
Name: Area, dtype: object
```

4. Children

The data type for the Children feature is numerical. The children variable will be used as an independent variable.

```
Summary statistics:  
count    10000.0000  
mean      2.0877  
std       2.1472  
min       0.0000  
25%       0.0000  
50%       1.0000  
75%       3.0000  
max       10.0000  
Name: Children, dtype: float64
```

5. Age

The data type for the Age feature is numerical. The Age variable will be used as an independent variable.

```
Summary statistics:  
count    10000.000000  
mean      53.078400  
std       20.698882  
min       18.000000  
25%       35.000000  
50%       53.000000  
75%       71.000000  
max       89.000000  
Name: Age, dtype: float64
```

6. Income

The data type for the income feature is numerical. The income variable will be used as an independent variable.

```
Summary statistics:
count      10000.000000
mean       39806.926771
std        28199.916702
min         348.670000
25%        19224.717500
50%        33170.605000
75%        53246.170000
max        258900.700000
Name: Income, dtype: float64
```

7. Marital

The data type for the Marital variable is categorical. The Marital variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Widowed' 'Married' 'Separated' 'Never Married' 'Divorced']
Summary statistics:
count      10000
unique         5
top      Divorced
freq       2092
Name: Marital, dtype: object
```

8. Gender

The data type for the Gender variable is categorical. The Gender variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Male' 'Female' 'Nonbinary']
Summary statistics:
count      10000
unique         3
top      Female
freq       5025
Name: Gender, dtype: object
```


9. Outage_sec_perweek

The data type for the Outage_sec_perweek feature is numerical. The Outage_sec_perweek variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      10.001848
std       2.976019
min       0.099747
25%       8.018214
50%      10.018560
75%      11.969485
max       21.207230
Name: Outage_sec_perweek, dtype: float64
```

10. Email

The data type for the Email feature is numerical. The Email variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      12.016000
std       3.025898
min       1.000000
25%      10.000000
50%      12.000000
75%      14.000000
max      23.000000
Name: Email, dtype: float64
```

11. Contacts

The data type for the contacts feature is numerical. The contact variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      0.994200
std       0.988466
min       0.000000
25%       0.000000
50%       1.000000
75%       2.000000
max       7.000000
Name: Contacts, dtype: float64
```

12. Yearly_equip_failure

The data type for the Yearly_equip_failure feature is numerical. The yearly_equip_failure variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      0.398000
std       0.635953
min       0.000000
25%       0.000000
50%       0.000000
75%       1.000000
max       6.000000
Name: Yearly_equip_failure, dtype: float64
```

13. Techie

The data type for the Techie variable is categorical. The Techie variable will be used as an independent variable. The variable will be converted to numerical data.

```
Summary statistics:  
count      10000  
unique       2  
top         No  
freq       8321  
Name: Techie, dtype: object
```

14. Contract

The data type for the Contract variable is categorical. The contract variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['One year' 'Month-to-month' 'Two Year']  
Summary statistics:  
count      10000  
unique       3  
top      Month-to-month  
freq       5456  
Name: Contract, dtype: object
```

15. Port_modem

The data type for the Port_modem variable is categorical. The Port_modem variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']  
Summary statistics:  
count      10000  
unique       2  
top         No  
freq       5166  
Name: Port_modem, dtype: object
```

16. Tablet

The data type for the Tablet variable is categorical. The Tablet variable will be used as an independent variable. The variable will be converted to numerical data.

```
Summary statistics:  
count      10000  
unique       2  
top         No  
freq       7009  
Name: Tablet, dtype: object
```

17. InternetService

The data type for the InternetService variable is categorical. The InternetService variable will be used as an independent variable. The variable will be converted to numerical data.

```
Summary statistics:  
count      10000  
unique       3  
top      Fiber Optic  
freq       4408  
Name: InternetService, dtype: object
```

18. Phone

The data type for the Phone variable is categorical. The Phone variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']
Summary statistics:
count      10000
unique       2
top        Yes
freq       9067
Name: Phone, dtype: object
```

19. Multiple

The data type for the Multiple variables is categorical. Multiple variables will be used as independent variables. The variable will be converted to numerical data.

```
Unique values: ['No' 'Yes']
Summary statistics:
count      10000
unique       2
top        No
freq       5392
Name: Multiple, dtype: object
```

20. OnlineSecurity

The data type for the Online Security variable is categorical. The Online Security variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']
Summary statistics:
count      10000
unique       2
top        No
freq       6424
Name: OnlineSecurity, dtype: object
```

21. OnlineBackup

The data type for the Online Backup variable is categorical. The Online Backup variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']
Summary statistics:
count      10000
unique       2
top         No
freq       5494
Name: OnlineBackup, dtype: object
```

22. DeviceProtection

The data type for the Device Protection variable is categorical. The Device Protection variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['No' 'Yes']
Summary statistics:
count      10000
unique       2
top         No
freq       5614
Name: DeviceProtection, dtype: object
```

23. TechSupport

The data type for the TechSupport variable is categorical. The TechSupport variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['No' 'Yes']
Summary statistics:
count      10000
unique       2
top         No
freq       6250
Name: TechSupport, dtype: object
```

24. StreamingTV

The data type for the StreamingTV variable is categorical. The StreamingTV variable will be used as an independent variable.

```
Unique values: ['No' 'Yes']
Summary statistics:
count      10000
unique       2
top         No
freq       5071
Name: StreamingTV, dtype: object
```

25. StreamingMovies

The data type for the Streaming Movies variable is categorical. The Streaming Movies variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']
Summary statistics:
count      10000
unique       2
top         No
freq       5110
Name: StreamingMovies, dtype: object
```

26. PaperlessBilling

The data type for the Paperless Billing variable is categorical. The PaperlessBilling variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Yes' 'No']
Summary statistics:
count      10000
unique       2
top         Yes
freq       5882
Name: PaperlessBilling, dtype: object
```

27. PaymentMethod

The data type for the PaymentMethod variable is categorical. The Payment Method variable will be used as an independent variable. The variable will be converted to numerical data.

```
Unique values: ['Credit Card (automatic)' 'Bank Transfer(automatic)' 'Mailed Check'
'Electronic Check']
Summary statistics:
count      10000
unique       4
top      Electronic Check
freq       3398
Name: PaymentMethod, dtype: object
```

28. Population

The data type for the Population feature is numerical. The population variable will be used as an independent variable.

```
Summary statistics:
count      10000.000000
mean       9756.562400
std        14432.698671
min         0.000000
25%        738.000000
50%        2910.500000
75%        13168.000000
max        111850.000000
Name: Population, dtype: float64
```

29. TimelyResponse

The data type for the TimelyResponse variable is numerical. The TimelyResponse variable will be used as an independent variable.

```
Summary statistics:
count      10000.000000
mean        3.490800
std         1.037797
min         1.000000
25%         3.000000
50%         3.000000
75%         4.000000
max         7.000000
Name: TimelyResponse, dtype: float64
```

30. Fixes

The data type for the Fixes variable is numerical. The Fixes variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.505100
std       1.034641
min       1.000000
25%       3.000000
50%       4.000000
75%       4.000000
max       7.000000
Name: Fixes, dtype: float64
```

31. Replacements

The data type for the Replacements variable is numerical. The Replacements variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.487000
std       1.027977
min       1.000000
25%       3.000000
50%       3.000000
75%       4.000000
max       8.000000
Name: Replacements, dtype: float64
```

32. Reliability

The data type for the Reliability variable is numerical. The Reliability variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.497500
std       1.025816
min       1.000000
25%       3.000000
50%       3.000000
75%       4.000000
max       7.000000
Name: Reliability, dtype: float64
```

33. Options

The data type for the Options variable is numerical. The Options variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.492900
std       1.024819
min       1.000000
25%       3.000000
50%       3.000000
75%       4.000000
max       7.000000
Name: Options, dtype: float64
```

34. Respectfulness

The data type for the Respectfulness variable is numerical. The Respectfulness variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.497300
std       1.033586
min       1.000000
25%       3.000000
50%       3.000000
75%       4.000000
max       8.000000
Name: Respectfulness, dtype: float64
```

35. Courteous

The data type for the Courteous variable is numerical. The Courteous variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.509500
std       1.028502
min       1.000000
25%       3.000000
50%       4.000000
75%       4.000000
max       7.000000
Name: Courteous, dtype: float64
```

36. Listening

The data type for the Listening variable is numerical. The Listening variable will be used as an independent variable.

```
Summary statistics:
count    10000.000000
mean      3.495600
std       1.028633
min       1.000000
25%       3.000000
50%       3.000000
75%       4.000000
max       8.000000
Name: Listening, dtype: float64
```

Churn_df_encoded.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 45 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Population                               10000 non-null  int64
1   Children                                10000 non-null  int64
2   Age                                      10000 non-null  int64
3   Income                                  10000 non-null  float64
4   Outage_sec_perweek                      10000 non-null  float64
5   Email                                    10000 non-null  int64
6   Contacts                                10000 non-null  int64
7   Yearly_equip_failure                    10000 non-null  int64
8   MonthlyCharge                           10000 non-null  float64
9   TimelyResponse                          10000 non-null  int64
10  Fixes                                   10000 non-null  int64
11  Replacements                            10000 non-null  int64
12  Reliability                             10000 non-null  int64
13  Options                                 10000 non-null  int64
14  Respectfulness                          10000 non-null  int64
15  Courteous                              10000 non-null  int64
16  Listening                                10000 non-null  int64
17  Area_Suburban                           10000 non-null  int64
18  Area_Urban                              10000 non-null  int64
19  Marital_Married                         10000 non-null  int64
20  Marital_Never Married                   10000 non-null  int64
21  Marital_Separated                       10000 non-null  int64
22  Marital_Widowed                         10000 non-null  int64
23  Gender_Male                             10000 non-null  int64
24  Gender_Nonbinary                        10000 non-null  int64
25  Churn_Yes                               10000 non-null  int64
26  Techie_Yes                              10000 non-null  int64
27  Contract_One year                       10000 non-null  int64
28  Contract_Two Year                        10000 non-null  int64
29  Port_modem_Yes                          10000 non-null  int64
30  Tablet_Yes                              10000 non-null  int64
31  InternetService_Fiber Optic             10000 non-null  int64
32  InternetService_Other                    10000 non-null  int64
33  Phone_Yes                               10000 non-null  int64
34  Multiple_Yes                            10000 non-null  int64
35  OnlineSecurity_Yes                      10000 non-null  int64
36  OnlineBackup_Yes                        10000 non-null  int64
37  DeviceProtection_Yes                    10000 non-null  int64
38  TechSupport_Yes                         10000 non-null  int64
39  StreamingTV_Yes                         10000 non-null  int64
40  StreamingMovies_Yes                     10000 non-null  int64
41  PaperlessBilling_Yes                    10000 non-null  int64
42  PaymentMethod_Credit Card (automatic)    10000 non-null  int64
43  PaymentMethod_Electronic Check           10000 non-null  int64
44  PaymentMethod_Mailed Check              10000 non-null  int64
dtypes: float64(3), int64(42)
memory usage: 3.4 MB
```

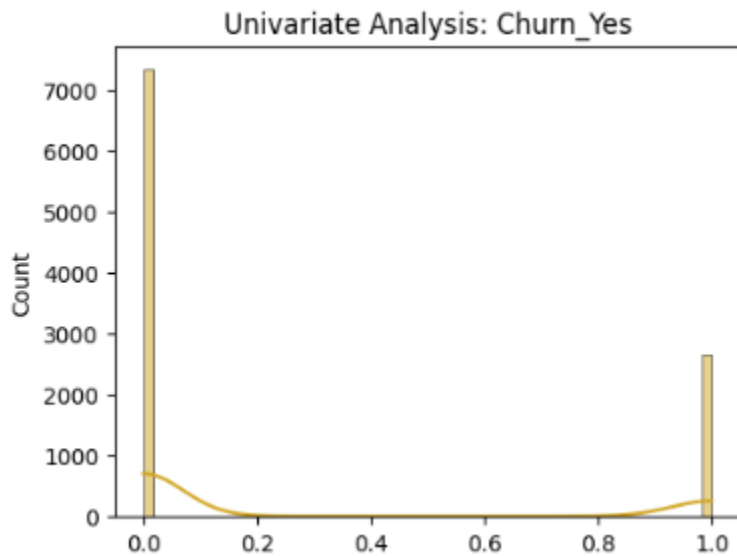
C3: VISUALIZATIONS

Univariate Analysis

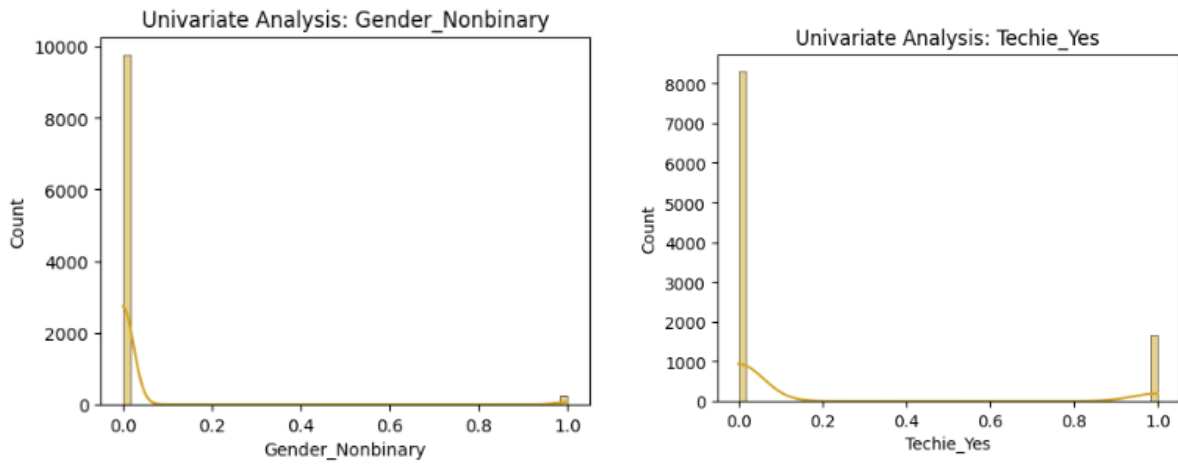
Bobbitt, Zach (2021, November 22) states that univariate analysis refers to the analysis of one variable.

In generating the univariate visualizations, the Churn_Yes has been utilized as the dependent variable and all the other variables as independent variables.

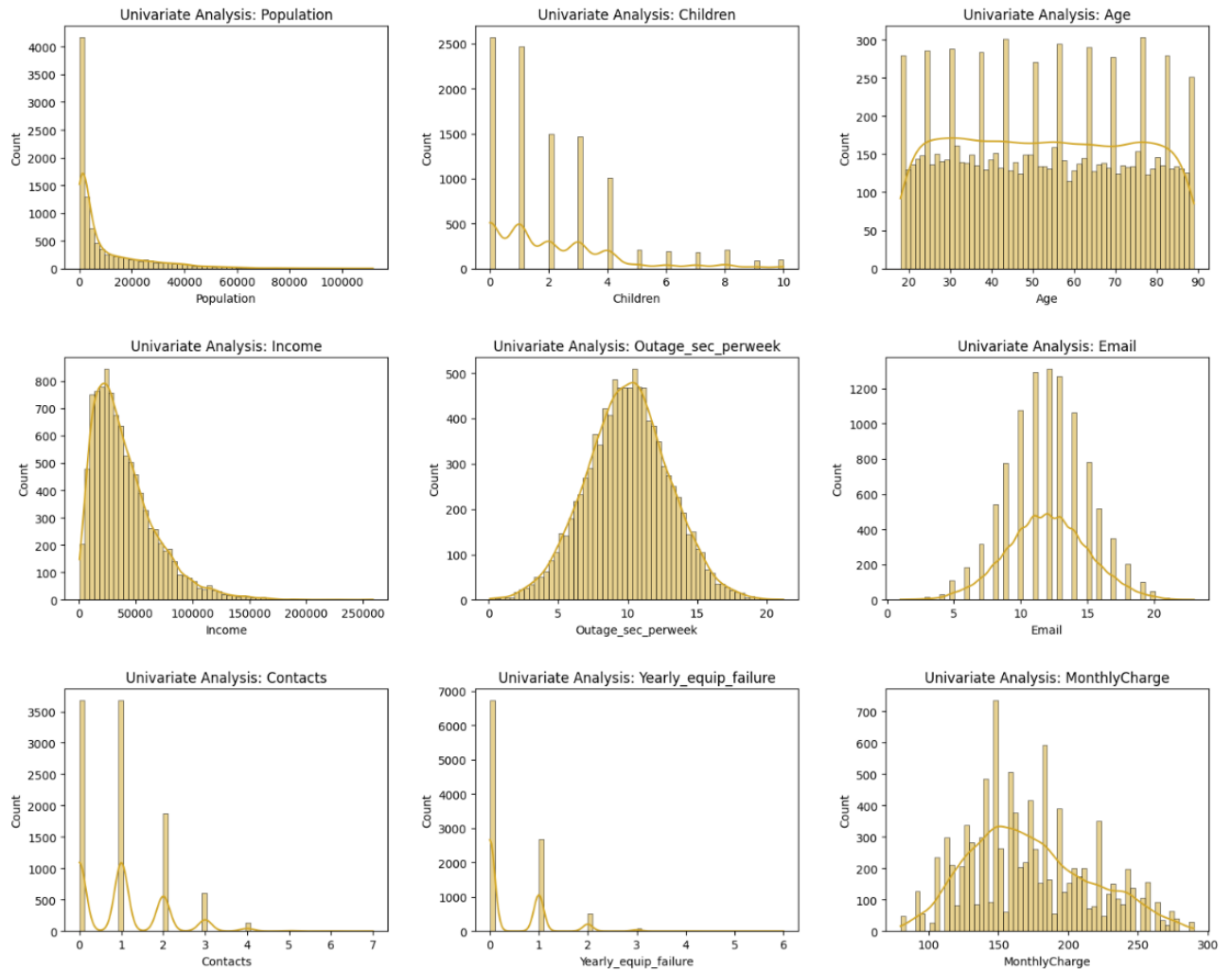
Dependent variable

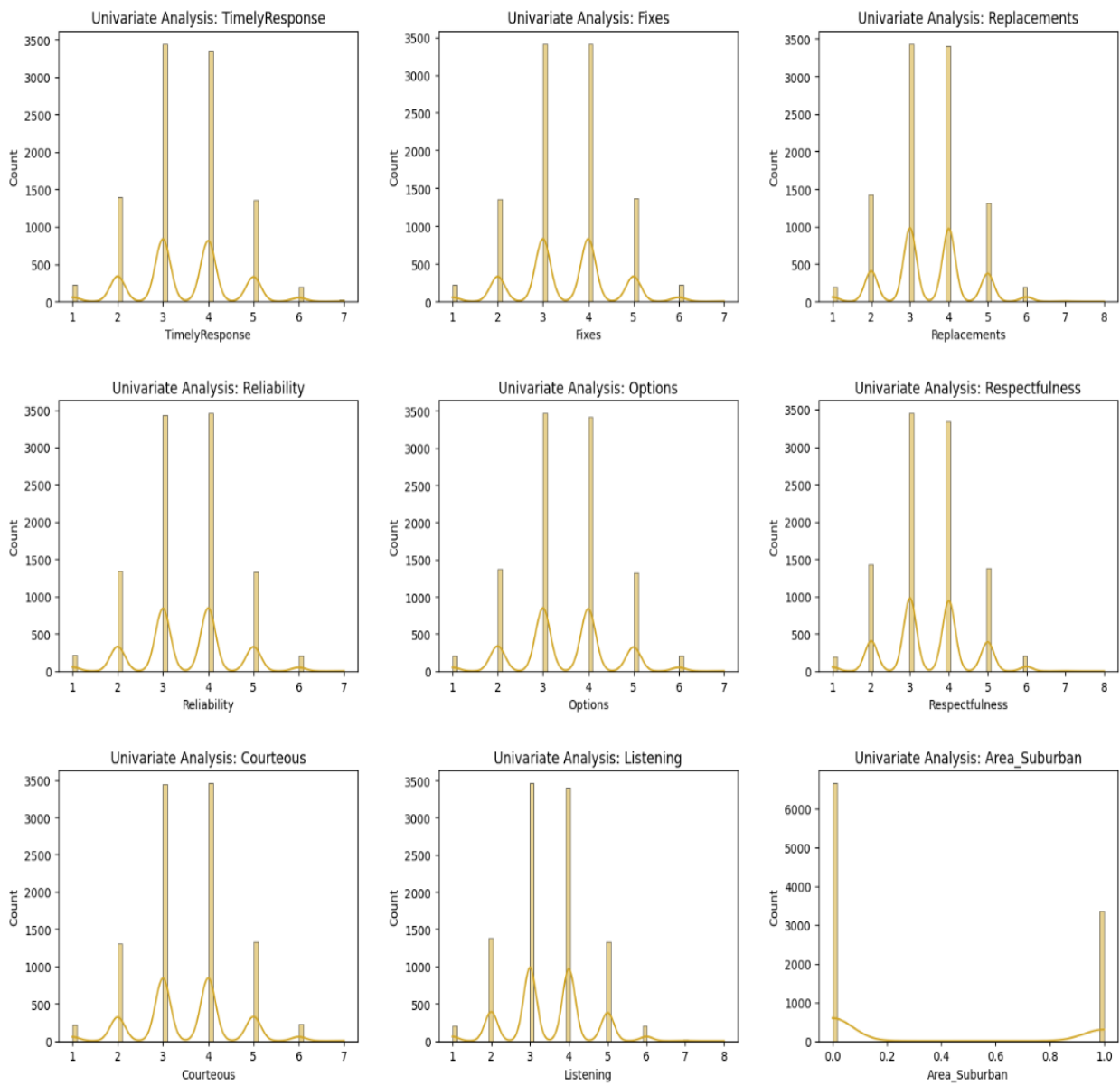


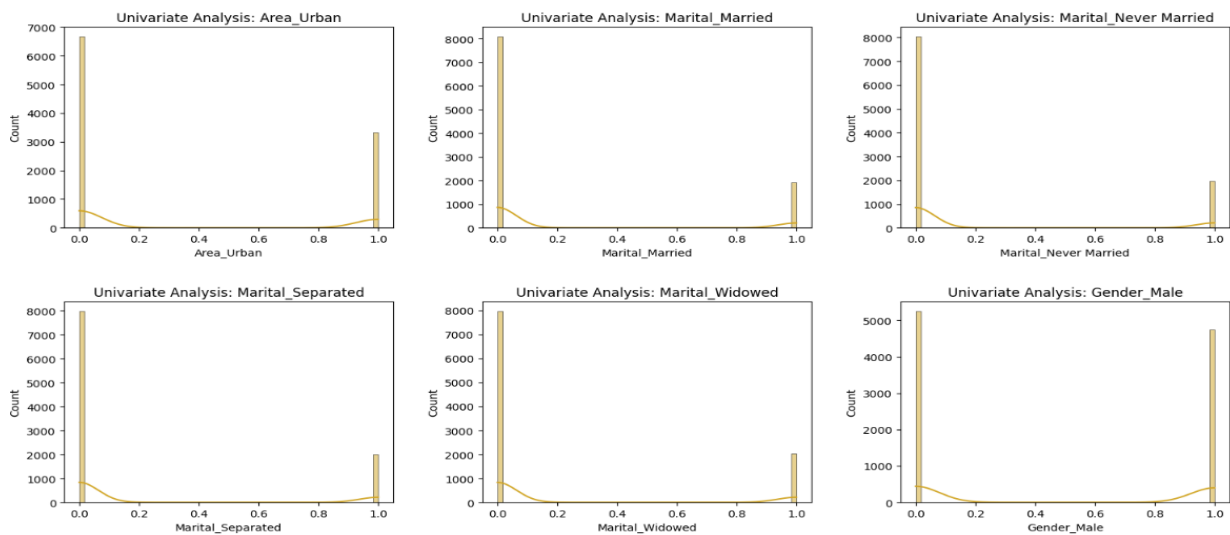
Independent variables



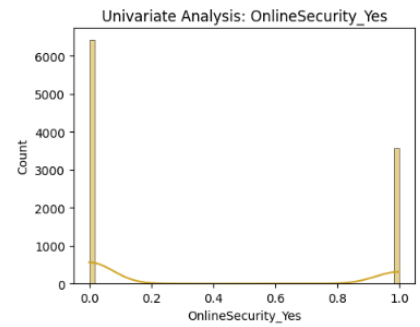
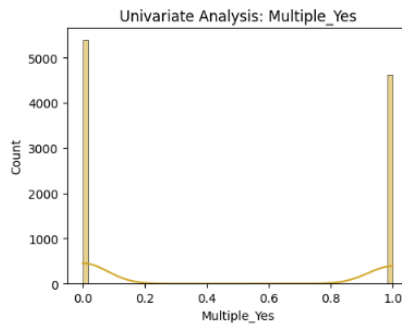
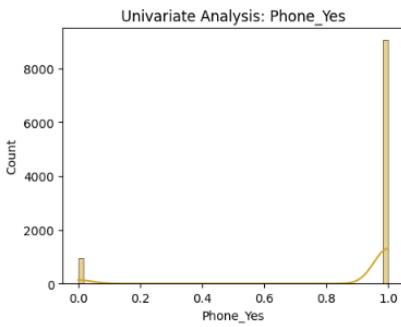
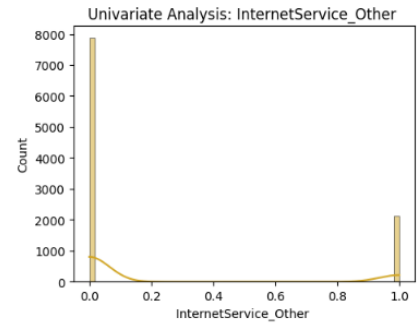
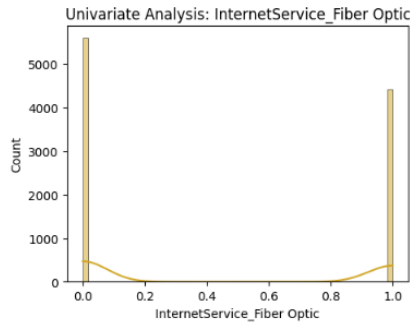
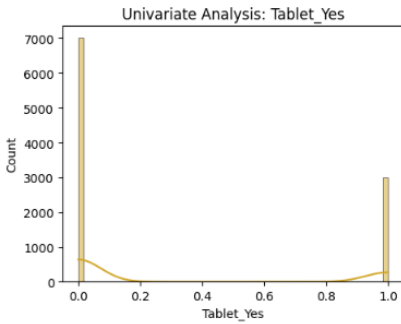
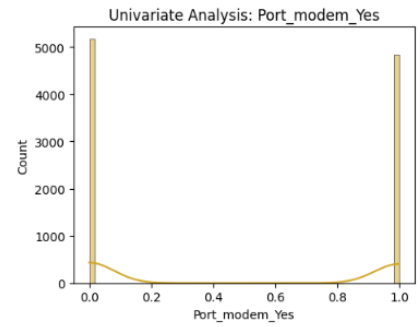
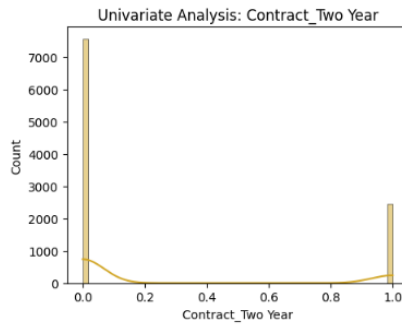
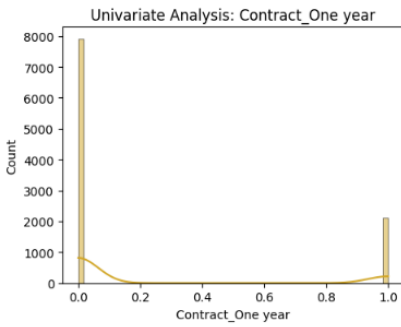
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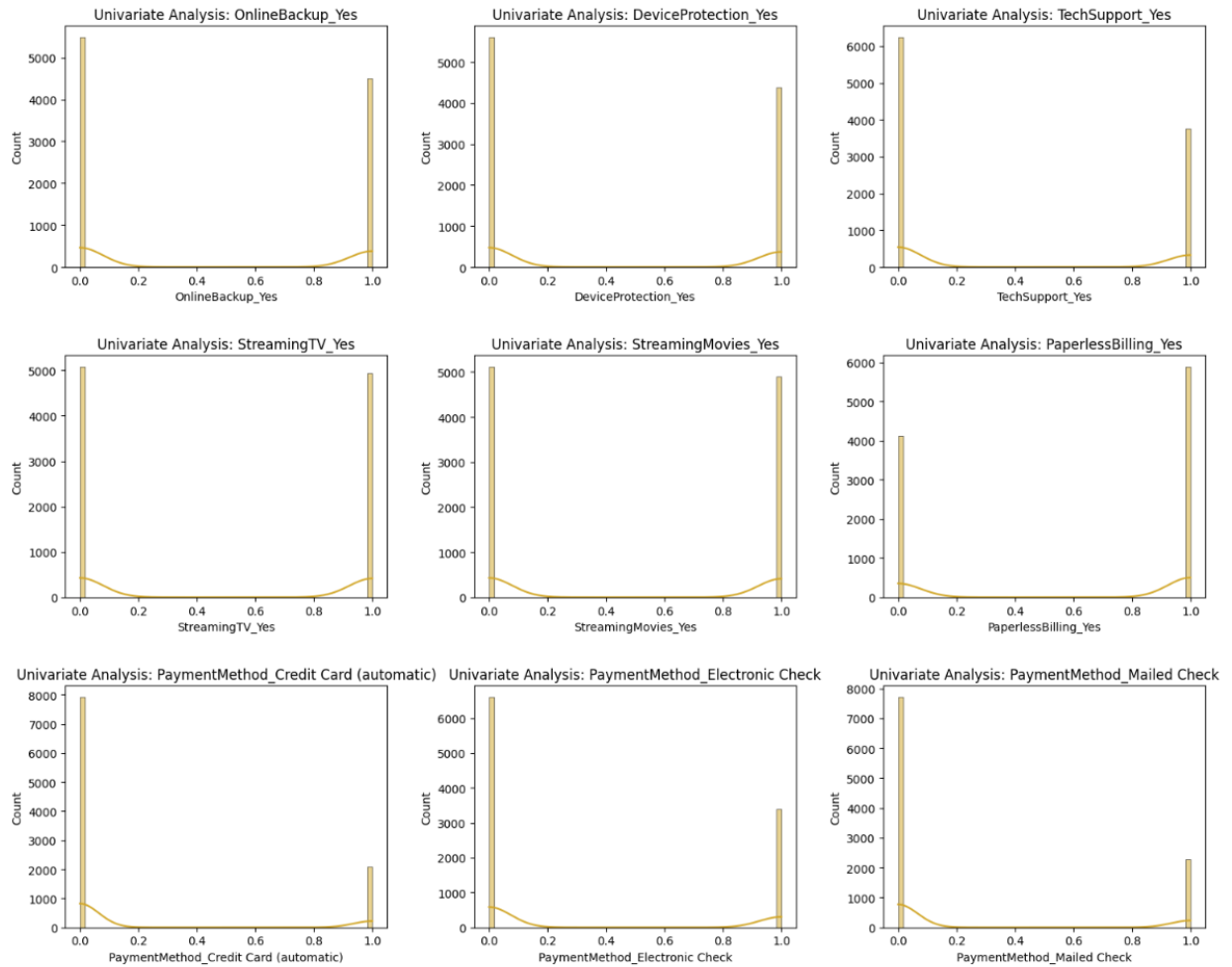






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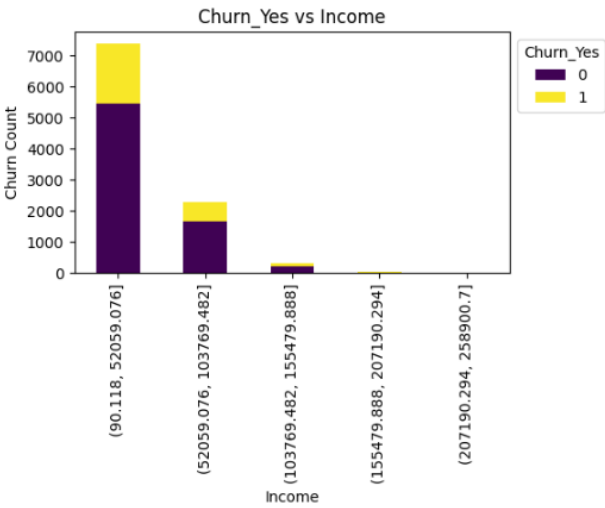
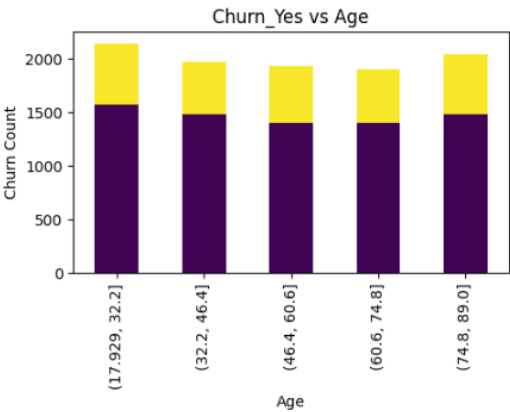
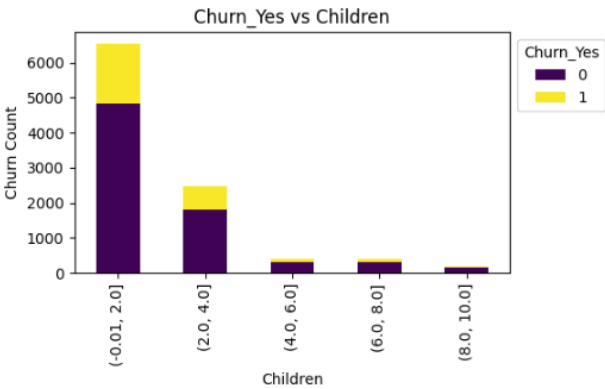
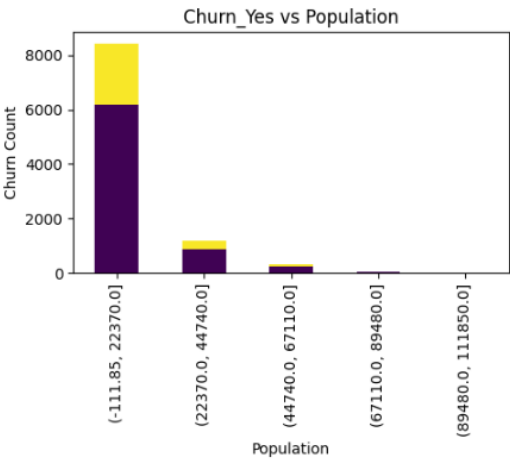


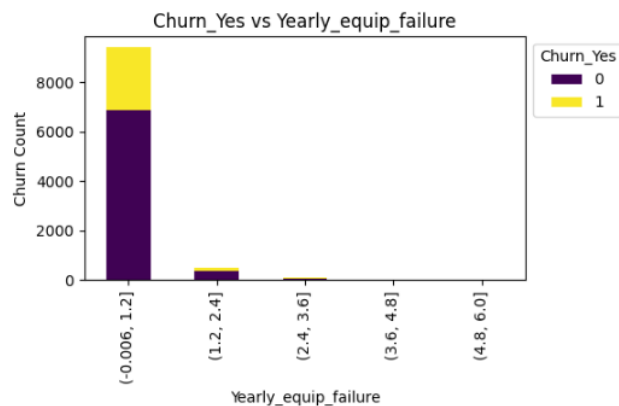
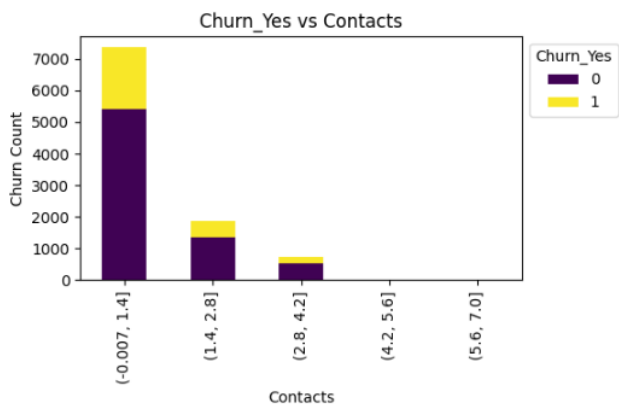
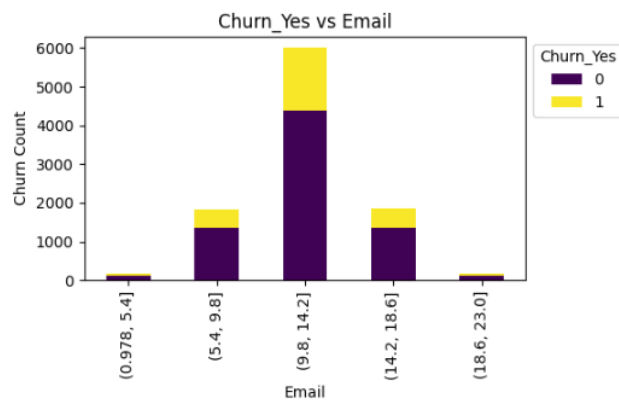
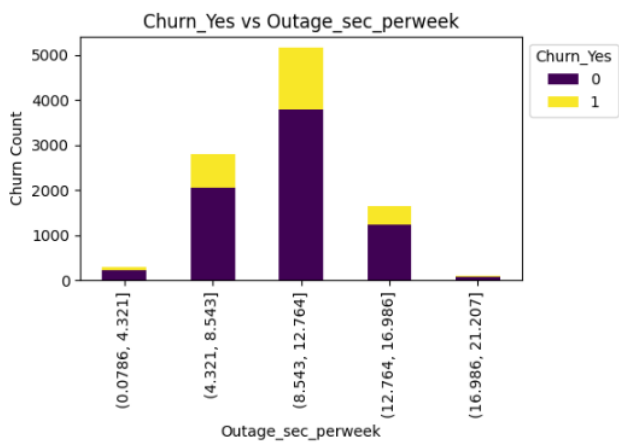


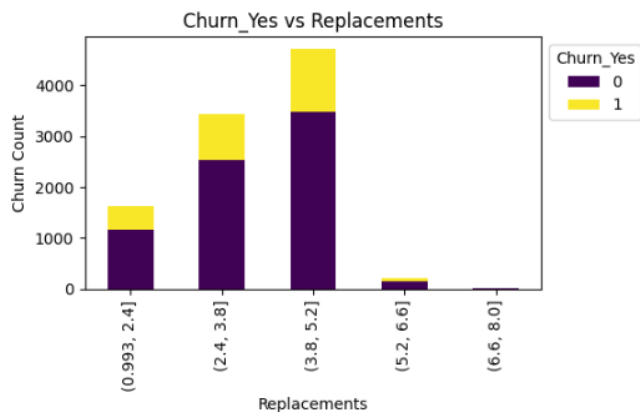
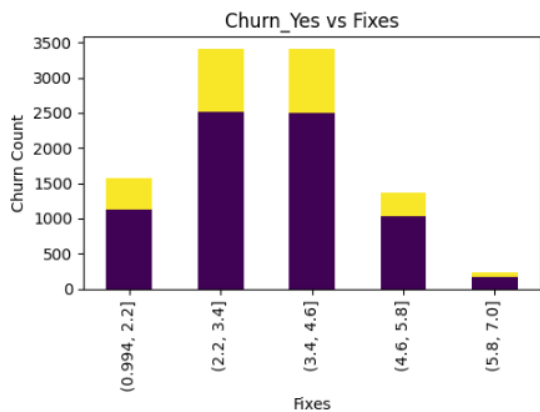
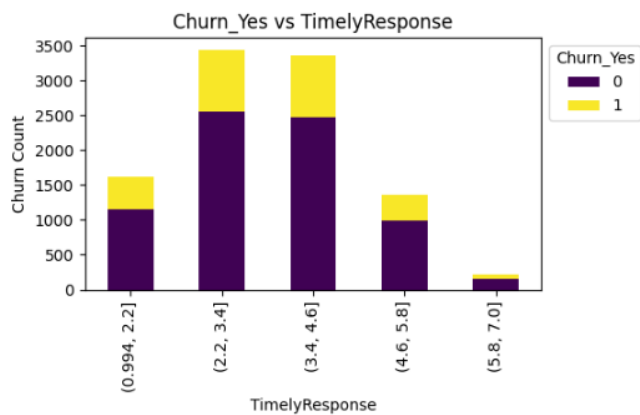
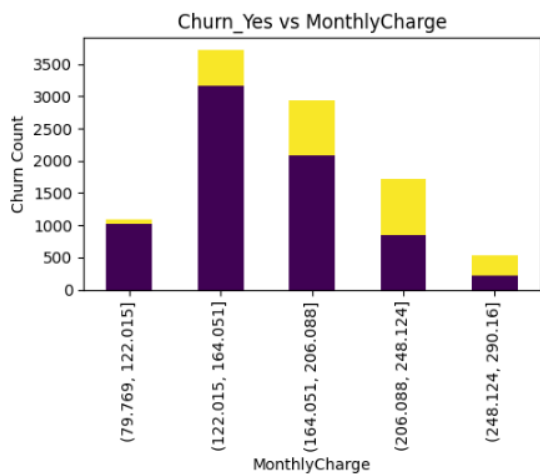
Bivariate analysis

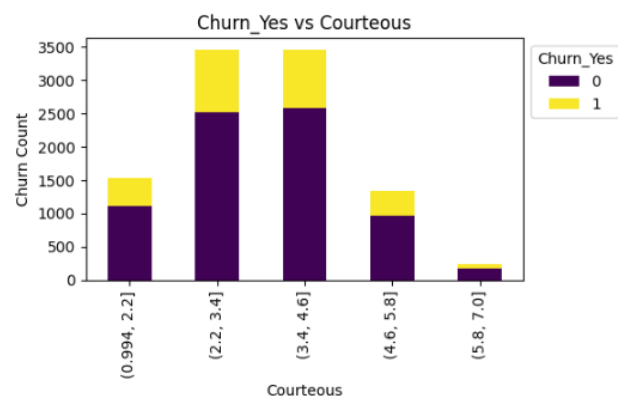
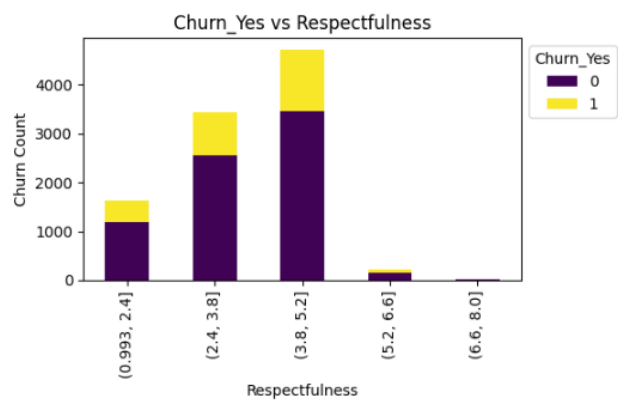
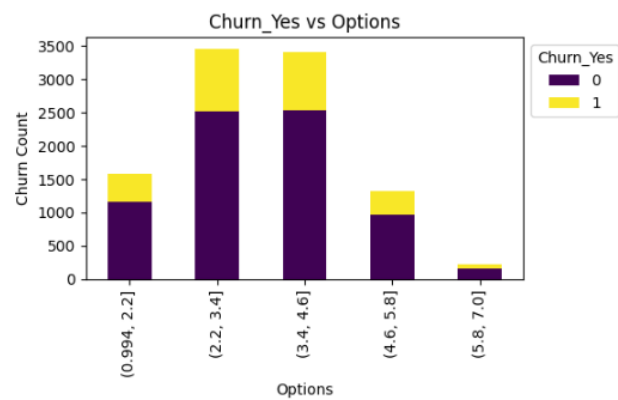
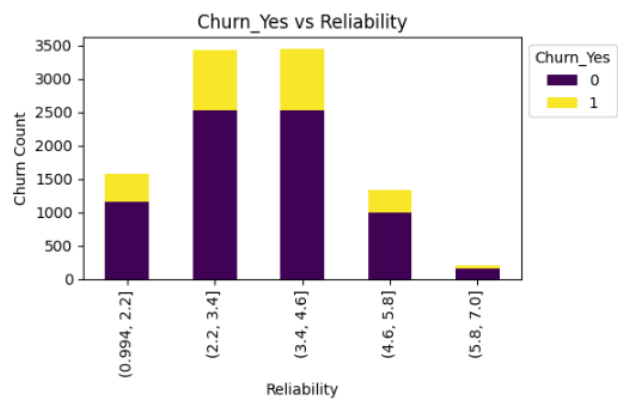
Bobbitt, Zach (2021, November 22) states that bivariate analysis refers to the analysis of two variables, and bivariate analysis aims to understand the relationship between two variables.

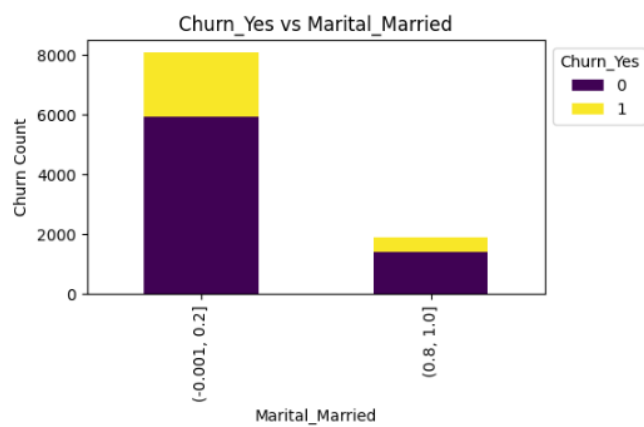
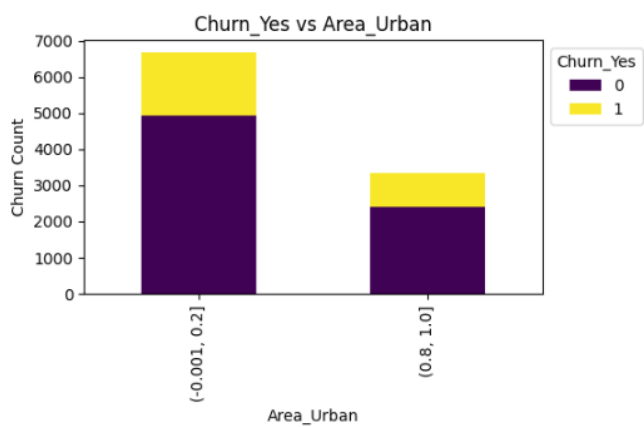
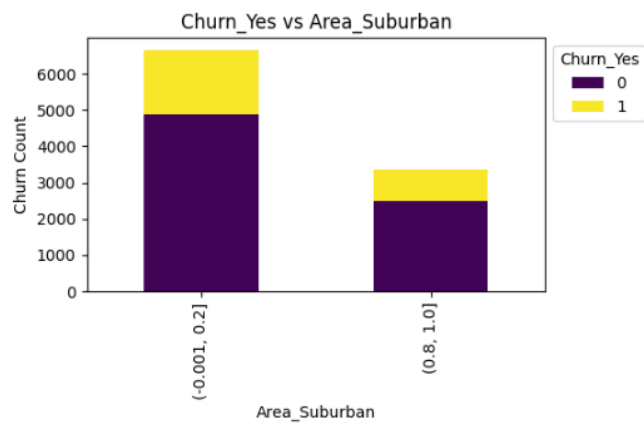
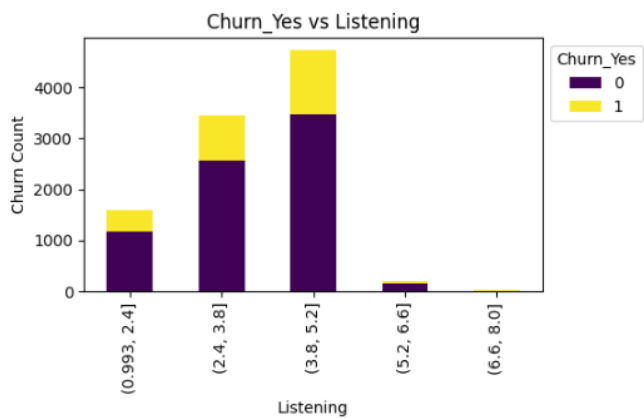
In generating the bivariate visualizations, the Churn_Yes has been utilized as the dependent variable and all the other variables as independent variables.

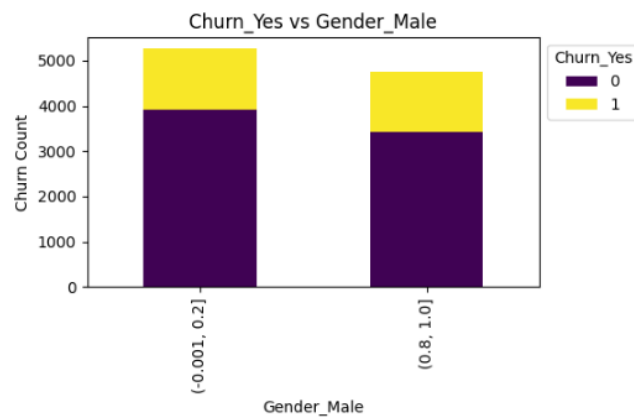
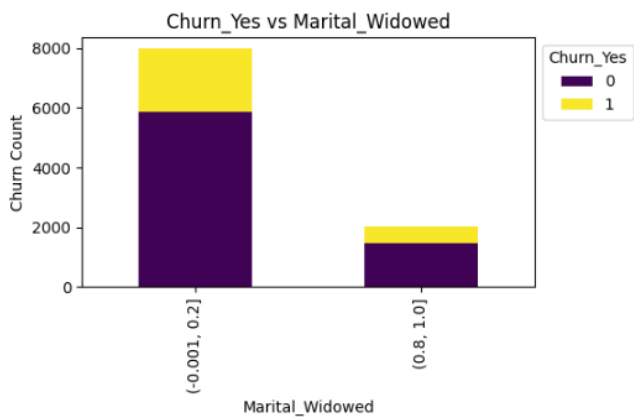
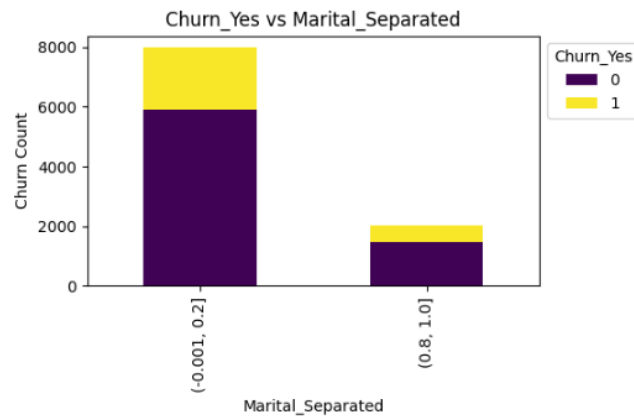
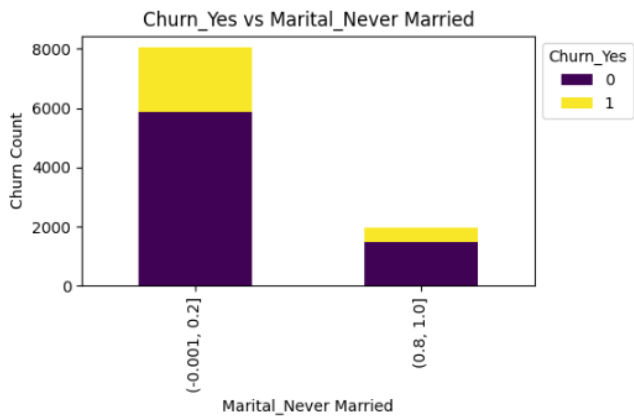


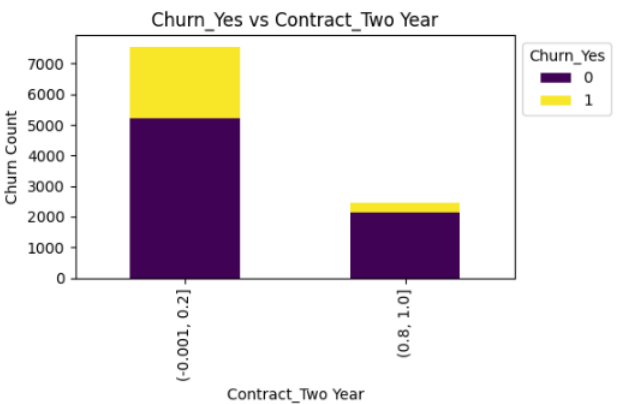
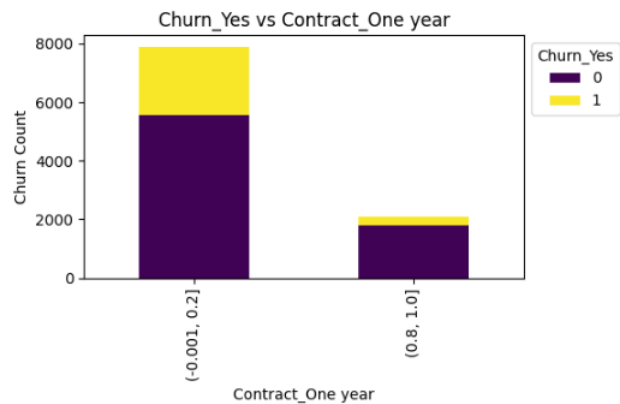
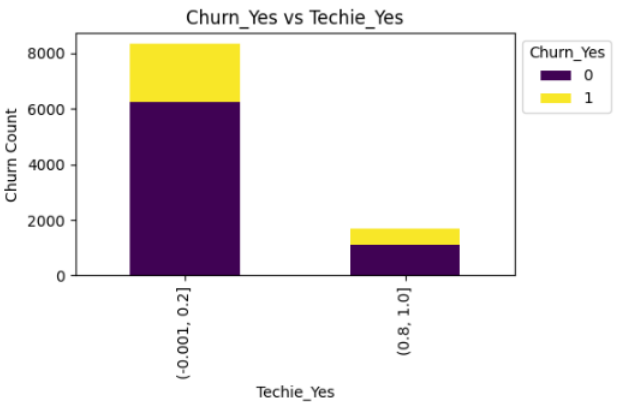
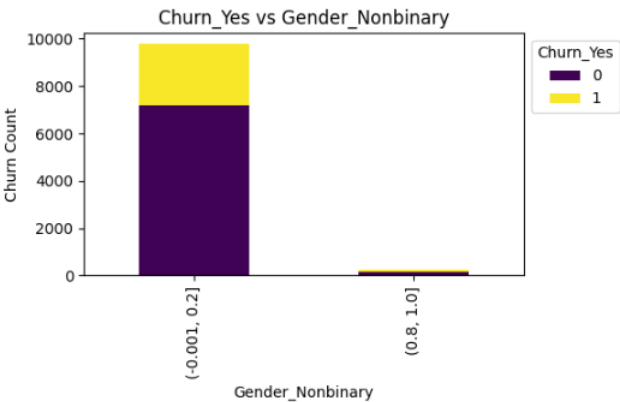


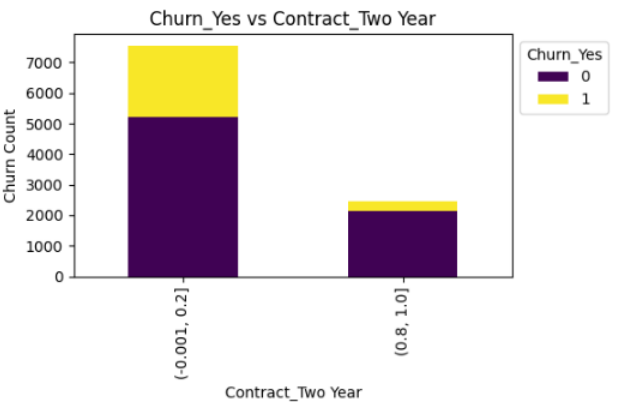
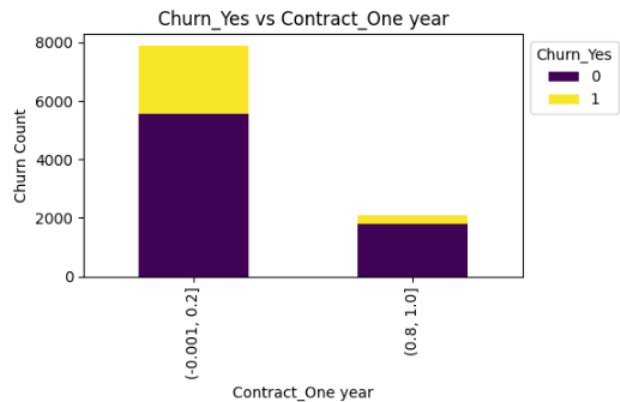
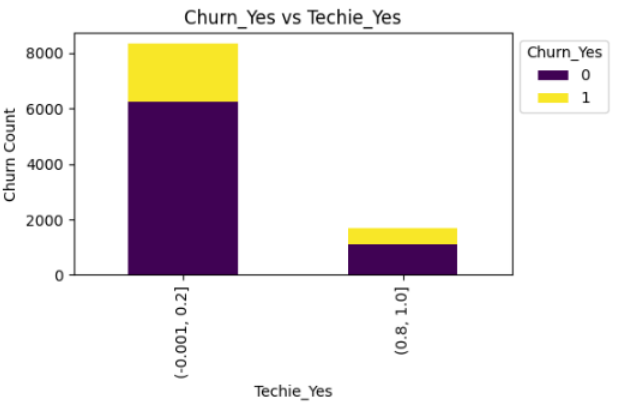
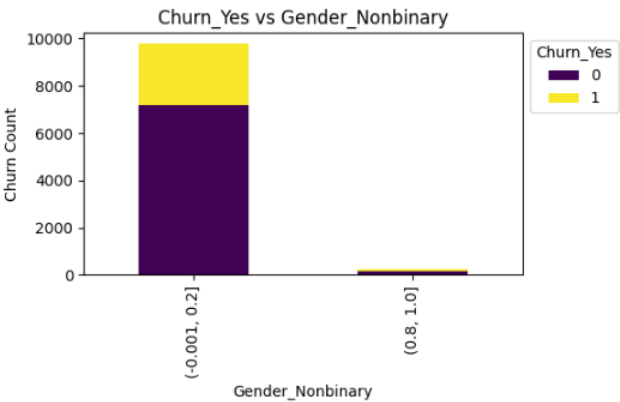


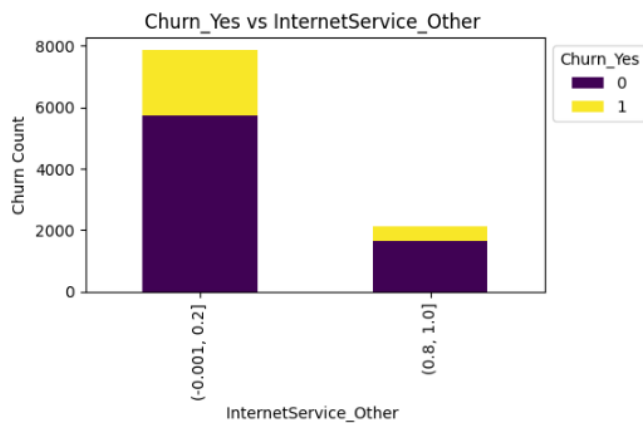
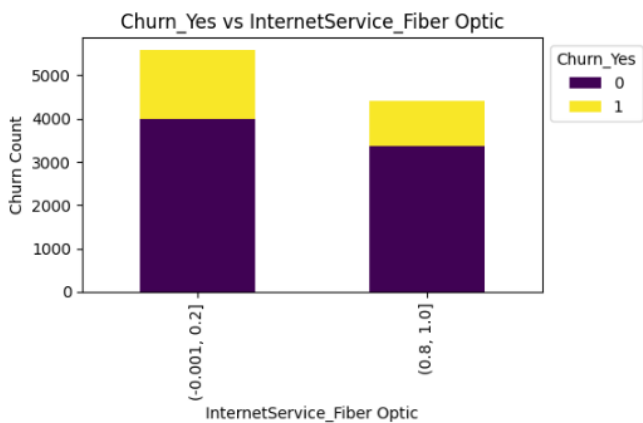
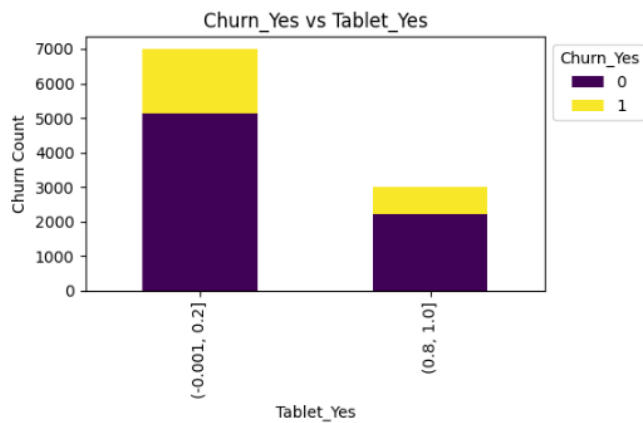
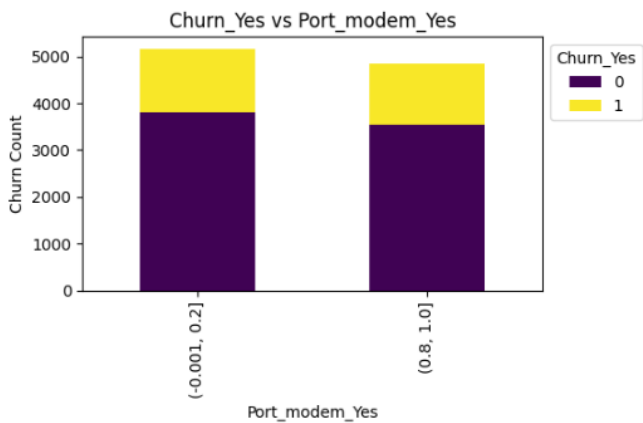


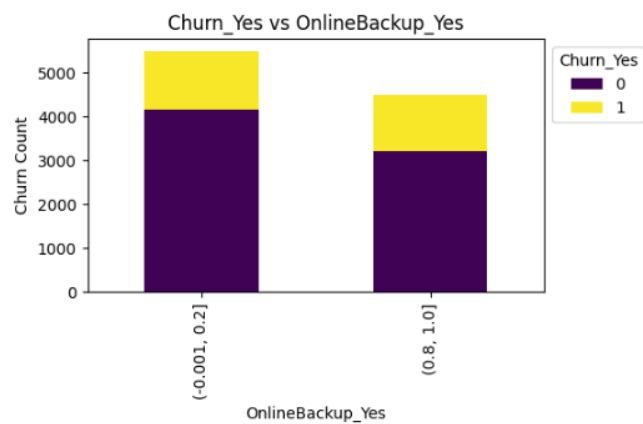
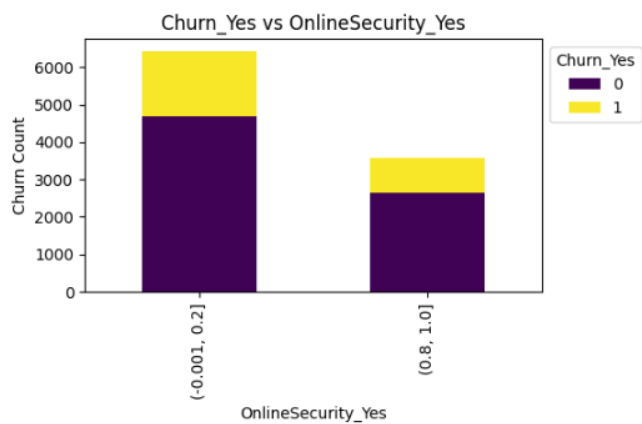
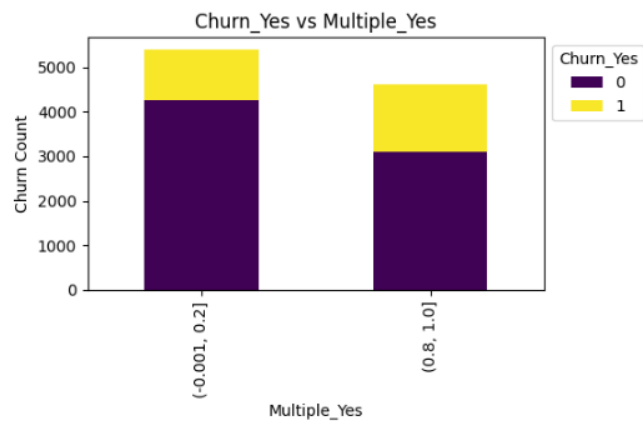
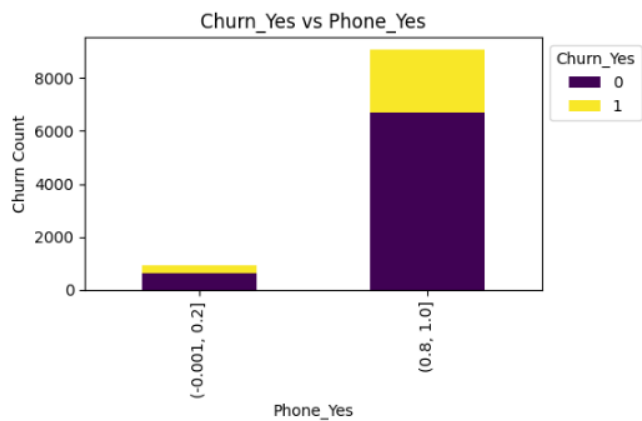


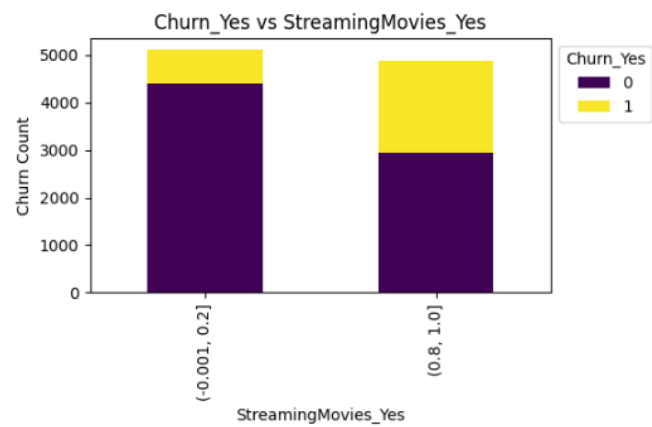
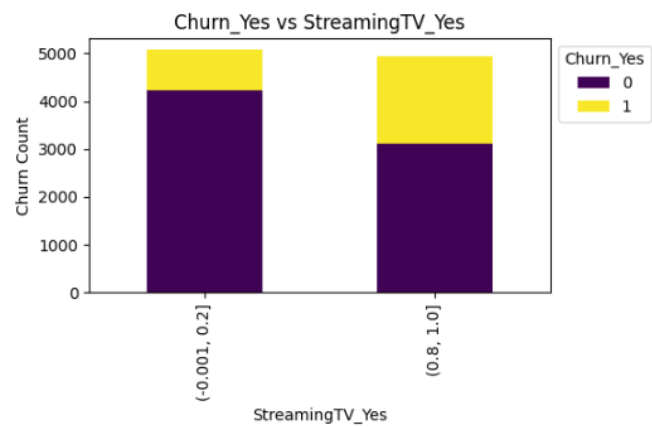
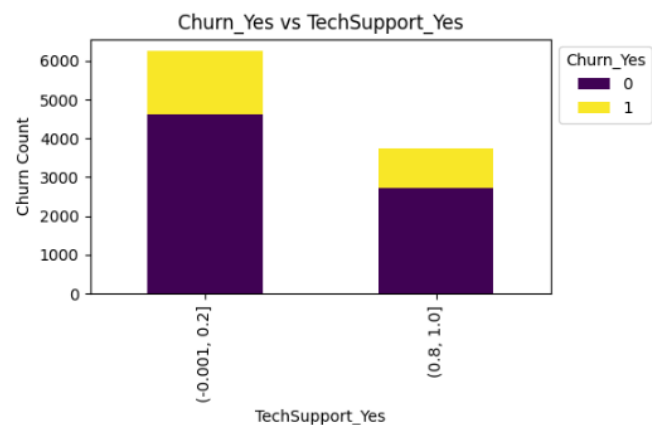
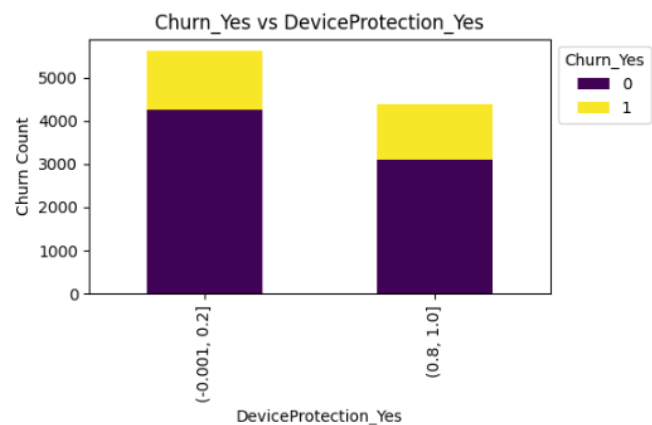


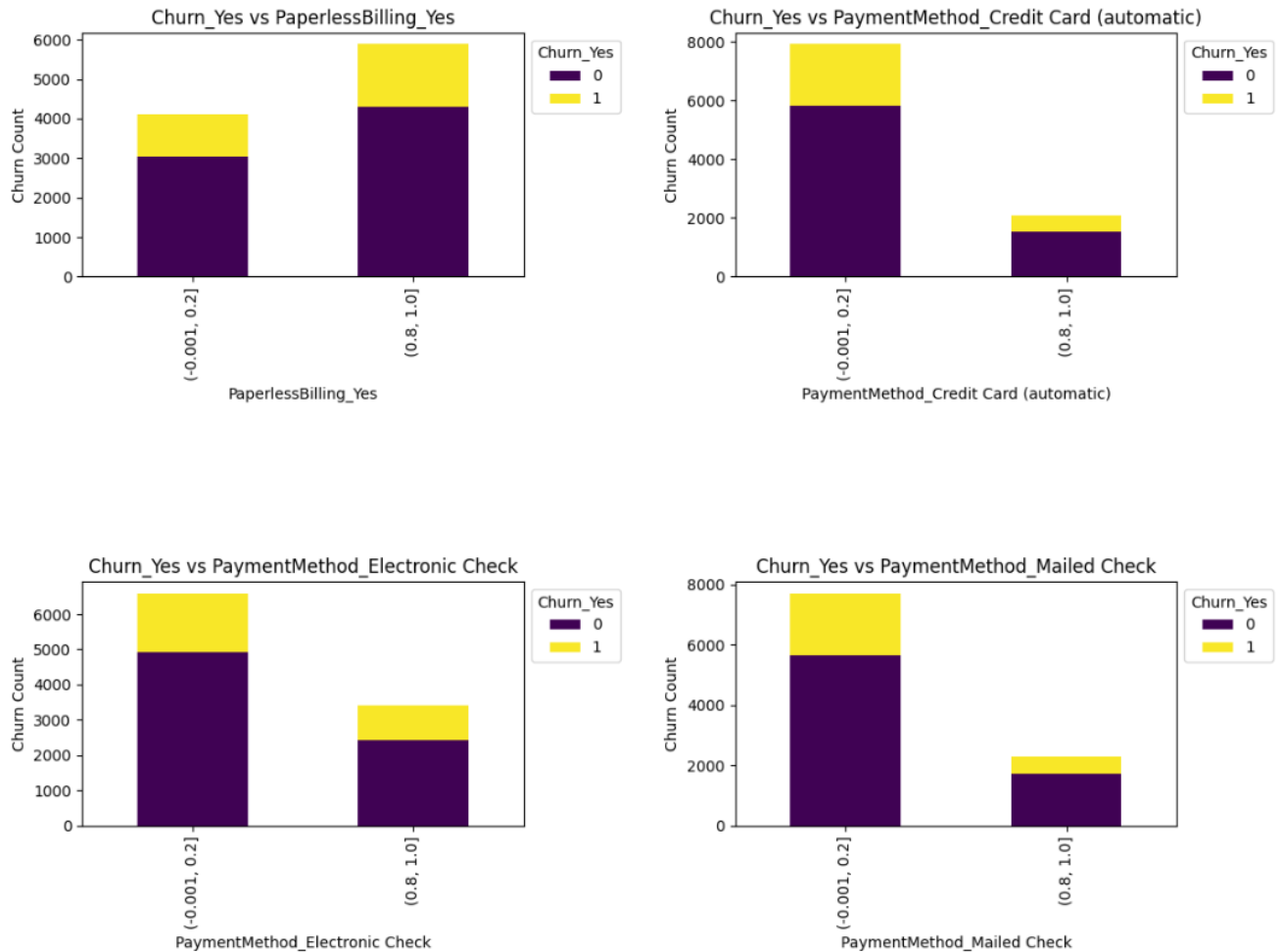












C4: DATA TRANSFORMATION

The data transformation involves dropping variables with high cardinality since all categorical variables will be covered with numerical variables. This is because many variables can lead to overfitting of the data. Logistics regression models also require that the independent variables have numeric data. The Categorical data will be converted to numerical data through dummy encoding (One Hot encoding). Dummy encoding will be used since dummy encoding typically drops one of the binary columns to prevent multicollinearity by setting the option `drop_first` as `true`.

C5: PREPARED DATA SET



Churn_df_encoded
_Logistics.csv

D1: INITIAL MODEL

```
Optimization terminated successfully.  
Current function value: 0.431173  
Iterations 7
```

Initial Model

Logit Regression Results

Dep. Variable:	Churn_Yes	No. Observations:	10000
Model:	Logit	Df Residuals:	9955
Method:	MLE	Df Model:	44
Date:	Thu, 21 Nov 2024	Pseudo R-squ.:	0.2543
Time:	07:26:11	Log-Likelihood:	-4311.7
Converged:	True	LL-Null:	-5782.2
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std_err	z	P> z	[0.025	0.975]
Population	-1.786e-07	1.88e-06	-0.095	0.924	-3.87e-06	3.51e-06
Children	0.0112	0.013	0.893	0.372	-0.013	0.036
Age	-0.0003	0.001	-0.210	0.833	-0.003	0.002
Income	6.186e-07	9.48e-07	0.653	0.514	-1.24e-06	2.48e-06
Outage_sec_perweek	-0.0066	0.009	-0.736	0.462	-0.024	0.011
Email	0.0125	0.009	1.404	0.160	-0.005	0.030
Contacts	0.0283	0.027	1.047	0.295	-0.025	0.081
Yearly equip_failure	-0.0717	0.043	-1.674	0.094	-0.156	0.012
MonthlyCharge	0.0091	0.003	2.774	0.006	0.003	0.016
TimelyResponse	0.0191	0.038	0.499	0.618	-0.056	0.094
Fixes	-0.0544	0.036	-1.506	0.132	-0.125	0.016
Replacements	-0.0022	0.033	-0.068	0.946	-0.067	0.062
Reliability	-0.0214	0.030	-0.721	0.471	-0.080	0.037
Options	-0.0400	0.031	-1.300	0.194	-0.100	0.020
Respectfulness	0.0137	0.031	0.436	0.663	-0.048	0.075
Courteous	-0.0090	0.030	-0.302	0.763	-0.067	0.049
Listening	0.0148	0.028	0.522	0.602	-0.041	0.070
Area_Suburban	-0.0203	0.066	-0.307	0.759	-0.150	0.109
Area_Urban	0.0903	0.066	1.376	0.169	-0.038	0.219
Marital_Married	-0.0040	0.085	-0.046	0.963	-0.171	0.163
Marital_Never Married	-0.0331	0.086	-0.387	0.699	-0.201	0.135
Marital_Separated	0.1112	0.084	1.330	0.183	-0.053	0.275
Marital_Widowed	0.1619	0.083	1.938	0.053	-0.002	0.326
Gender_Male	0.1545	0.054	2.840	0.005	0.048	0.261
Gender_Nonbinary	-0.0222	0.189	-0.117	0.907	-0.392	0.348
Techie_Yes	0.5238	0.069	7.596	0.000	0.389	0.659
Contract_One year	-1.6962	0.079	-21.561	0.000	-1.850	-1.542
Contract_Two Year	-1.8475	0.077	-23.906	0.000	-1.999	-1.696
Port_modem_Yes	0.0385	0.054	0.717	0.473	-0.067	0.144
Tablet_Yes	-0.0456	0.059	-0.776	0.438	-0.161	0.070
InternetService_Fiber Optic	-0.7899	0.090	-8.816	0.000	-0.965	-0.614
InternetService_Other	-0.4906	0.085	-5.741	0.000	-0.658	-0.323
Phone_Yes	-0.1336	0.090	-1.480	0.139	-0.310	0.043
Multiple_Yes	0.5686	0.119	4.760	0.000	0.334	0.803
OnlineSecurity_Yes	-0.1303	0.057	-2.285	0.022	-0.242	-0.019
OnlineBackup_Yes	0.1272	0.092	1.386	0.166	-0.053	0.307
DeviceProtection_Yes	0.1998	0.068	2.944	0.003	0.067	0.333
TechSupport_Yes	0.0380	0.069	0.551	0.581	-0.097	0.173
StreamingTV_Yes	1.0459	0.167	6.265	0.000	0.719	1.373
StreamingMovies_Yes	1.2478	0.196	6.370	0.000	0.864	1.632
PaperlessBilling_Yes	0.0804	0.055	1.472	0.141	-0.027	0.187
PaymentMethod_Credit Card (automatic)	0.0715	0.083	0.863	0.388	-0.091	0.234
PaymentMethod_Electronic Check	0.2519	0.074	3.416	0.001	0.107	0.396
PaymentMethod_Mailed Check	0.0854	0.081	1.056	0.291	-0.073	0.244
const	-3.4045	0.411	-8.277	0.000	-4.211	-2.598

D2: JUSTIFICATION OF MODEL REDUCTION

In determining the features to drop, the features that have a p-value >0.05 . The p-value represents the probability of observing something. A high p-value indicates that the results are not statistically significant. Beers, Brian (2024, November 13) states that a p-value, or probability value, is a number describing the likelihood of obtaining the observed data under the null hypothesis of a statistical test. The p-value is an alternative to rejection points to provide the smallest significance level at which the null hypothesis would be rejected. A smaller p-value means stronger evidence in favor of the alternative hypothesis.

The independent features below will be dropped since they have a high p-value, which means they are not statistically significant in determining the dependent variable Churn.

In building the model reduction, the following features were dropped due to a p-value greater than 0.05:

1. Marital_Married with a p-value of 0.962945.
2. Replacements with a p-value of 0.945737.
3. Population with a p-value of 0.924263.
4. Gender_Nonbinary with a p-value of 0.907632.
5. Age with a p-value of 0.835587.
6. Courteous with a p-value of 0.760418.
7. Area_Suburban with a p-value of 0.756041.
8. Respectfulness with a p-value of 0.700614.
9. Marital_Never Married with a p-value of 0.679702.
10. Listening with a p-value of 0.584231.
11. TechSupport_Yes with a p-value of 0.582231.

12. Income with a p-value of 0.519898.
13. Reliability with a p-value of 0.503023.
14. TimelyResponse with a p-value of 0.492727.
15. Port_modem_Yes with a p-value of 0.471232.
16. Outage_sec_perweek with a p-value of 0.461733.
17. Tablet_Yes with a p-value of 0.428761.
18. PaymentMethod_Credit Card (automatic) with a p-value of 0.404138.
19. PaymentMethod_Mailed Check with a p-value of 0.444117.
20. Children with a p-value of 0.356962.
21. Contacts with a p-value of 0.340726.
22. OnlineBackup_Yes with a p-value of 0.199939.
23. Options with a p-value of 0.199743.
24. Fixes with a p-value of 0.158081.
25. Email with a p-value of 0.150733.
26. PaperlessBilling_Yes with a p-value of 0.144263.
27. Phone_Yes with a p-value of 0.115821.
28. Yearly_equip_failure with a p-value of 0.106462.
29. Area_Urban with a p-value of 0.075049.
30. Marital_Separated with a p-value of 0.069156.

Dropped features due to p-value > 0.05:

- Marital_Married: p-value = 0.962945
- Replacements: p-value = 0.945737
- Population: p-value = 0.924263
- Gender_Nonbinary: p-value = 0.907632
- Age: p-value = 0.835587
- Courteous: p-value = 0.760418
- Area_Suburban: p-value = 0.756041
- Respectfulness: p-value = 0.700614
- Marital_Never Married: p-value = 0.679702
- Listening: p-value = 0.584231
- TechSupport_Yes: p-value = 0.582231
- Income: p-value = 0.519898
- Reliability: p-value = 0.503023
- TimelyResponse: p-value = 0.492727
- Port_modem_Yes: p-value = 0.471232
- Outage_sec_perweek: p-value = 0.461733
- Tablet_Yes: p-value = 0.428761
- PaymentMethod_Credit Card (automatic): p-value = 0.404138
- PaymentMethod_Mailed Check: p-value = 0.444117
- Children: p-value = 0.356962
- Contacts: p-value = 0.340726
- OnlineBackup_Yes: p-value = 0.199939
- Options: p-value = 0.199743
- Fixes: p-value = 0.158081
- Email: p-value = 0.150733
- PaperlessBilling_Yes: p-value = 0.144263
- Phone_Yes: p-value = 0.115821
- Yearly_equip_failure: p-value = 0.106462
- Area_Urban: p-value = 0.075049
- Marital_Separated: p-value = 0.069156

D3: REDUCED LOGISTIC REGRESSION MODEL

```

Reduced Logistic Regression Model Summary:
                        Logit Regression Results
=====
Dep. Variable:         Churn_Yes    No. Observations:         10000
Model:                 Logit       Df Residuals:                 9985
Method:                MLE         Df Model:                   14
Date:                  Thu, 21 Nov 2024    Pseudo R-squ.:             0.2519
Time:                  07:43:59    Log-Likelihood:            -4325.8
converged:              True        LL-Null:                   -5782.2
Covariance Type:       nonrobust    LLR p-value:                0.000
=====
                        coef      std err      z      P>|z|      [0.025      0.975]
-----
MonthlyCharge          0.0128      0.002      7.209      0.000      0.009      0.016
Marital_Widowed        0.1445      0.066      2.188      0.029      0.015      0.274
Gender_Male            0.1599      0.054      2.982      0.003      0.055      0.265
Techie_Yes            0.5198      0.069      7.566      0.000      0.385      0.654
Contract_One year     -1.6996      0.079     -21.650      0.000     -1.854     -1.546
Contract_Two Year     -1.8450      0.077     -23.928      0.000     -1.996     -1.694
InternetService_Fiber Optic
-0.8548      0.070     -12.175      0.000     -0.992     -0.717
InternetService_Other
-0.4411      0.077      -5.694      0.000     -0.593     -0.289
Multiple_Yes          0.4453      0.078      5.739      0.000      0.293      0.597
OnlineSecurity_Yes    -0.1380      0.056      -2.449      0.014     -0.249     -0.028
DeviceProtection_Yes  0.1618      0.058      2.785      0.005      0.048      0.276
StreamingTV_Yes       0.8662      0.099      8.749      0.000      0.672      1.060
StreamingMovies_Yes   1.0359      0.113      9.179      0.000      0.815      1.257
PaymentMethod_Electronic Check
0.1929      0.056      3.433      0.001      0.083      0.303
const                -3.7988      0.191     -19.917      0.000     -4.173     -3.425
=====

```

E1: MODEL COMPARISON

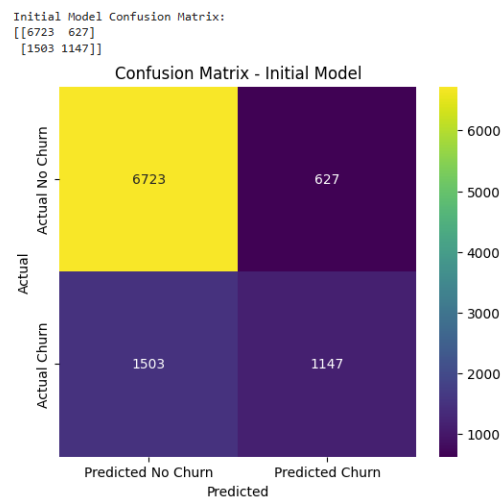
- a model evaluation metric

1. Confusion Matrix

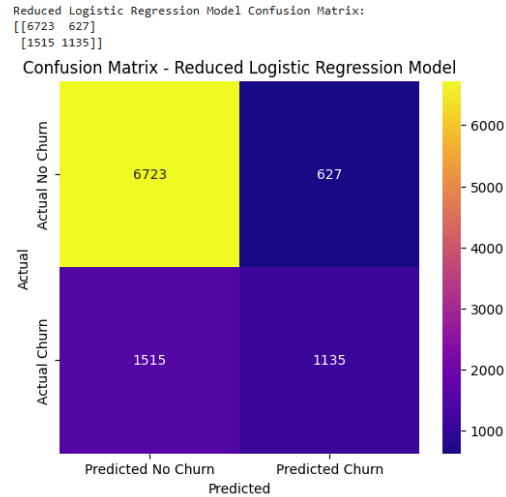
According to Narkhede, Sarang (2018, May 9), a confusion matrix is a performance measurement for machine learning classification problems where the output can be two or more classes. It is a table with four different combinations of predicted and actual values and is extremely useful for measuring Recall, Precision, Specificity, Accuracy, and AUC-ROC curves. A confusion matrix, also known as an error matrix or table of confusion, has four components: True positive, True negative, False Positive, and False negative. True Positive (TP). The model correctly predicted a positive outcome, and the actual outcome was positive. True Negative (TN). The model correctly predicted a

negative outcome, and the actual outcome was negative. False Positive (FP). The model incorrectly predicted a positive outcome, and the actual outcome was negative. This is type 1 error. False Negative (FN). The model incorrectly predicted a negative outcome, and the actual outcome was positive. This is a type II error

- initial logistic regression model



- reduced logistic regression model



The initial model had 1147 true positives and 1503 false negatives, while the reduced model had 1135 true positives and 1515 false negatives. This indicates a slight reduction in the number of true positives and a slight increase in false negatives after the reduction. True Negatives and False Positives remained unchanged, meaning the reduced model is equally good at correctly identifying negatives and making false positive predictions. The reduced model has a small bias toward predicting fewer positives, possibly at the cost of sensitivity; the change is minimal.

2. Accuracy

Accuracy is the ratio of correctly predicted instances to the total instances

- initial logistic regression model

Initial Model Accuracy: 0.787

- reduced logistic regression model

Reduced Logistic Regression Model Accuracy: 0.7858

There is no significant difference in the accuracy of the models since the initial model is at 78.7%, whereas the reduced model is at 78.6% in accuracy.

3. Classification report

- Precision is the ratio of true positive predictions to the total positive predictions made by the model.
 - Sensitivity is also called recall, the ratio of true positive predictions to the sum of true positive and false negative predictions.
 - F1-Score is the harmonic mean of precision and sensitivity.
 - Specificity is the ratio of true negative predictions to the sum of true negative and false positive predictions.
- initial logistic regression model

```
Initial Model Classification Report:
              precision    recall  f1-score   support

     0       0.82         0.91         0.86         7350
     1       0.65         0.43         0.52         2650

 accuracy          0.79         10000
 macro avg         0.73         0.67         0.69         10000
 weighted avg         0.77         0.79         0.77         10000
```

- reduced logistic regression model

Reduced Logistic Regression Model Classification Report:					
	precision	recall	f1-score	support	
0	0.82	0.91	0.86	7350	
1	0.64	0.43	0.51	2650	
accuracy			0.79	10000	
macro avg	0.73	0.67	0.69	10000	
weighted avg	0.77	0.79	0.77	10000	

- There is a small decrease in precision (0.65 to 0.64) in the reduced model. This minimal change suggests that the reduced model is still quite accurate in predicting positive cases but slightly less compared to the initial model.
- both models have sensitivity for class 0, being very high (0.91) and sensitivity for class 1, remaining low (0.43). The model does not change in its ability to identify negative or positive instances correctly.
- The F1-score for Class 0 remains unchanged at 0.86 in both models, indicating consistent performance in predicting negatives.
- Both models' macro average metrics are identical, indicating that the overall balance between precision, recall, and F1-score for both classes has not changed.
- The weighted average metrics are also unchanged between the initial and reduced models. This suggests that the reduced model doesn't significantly alter the overall performance despite small changes in precision and F1-score for Class 1.

4. AUC

AUC evaluates how well the model ranks the instances, making it more reliable for assessing performance across different thresholds and class distributions

- initial logistic regression model

Initial Model AUC: 0.832341932999615

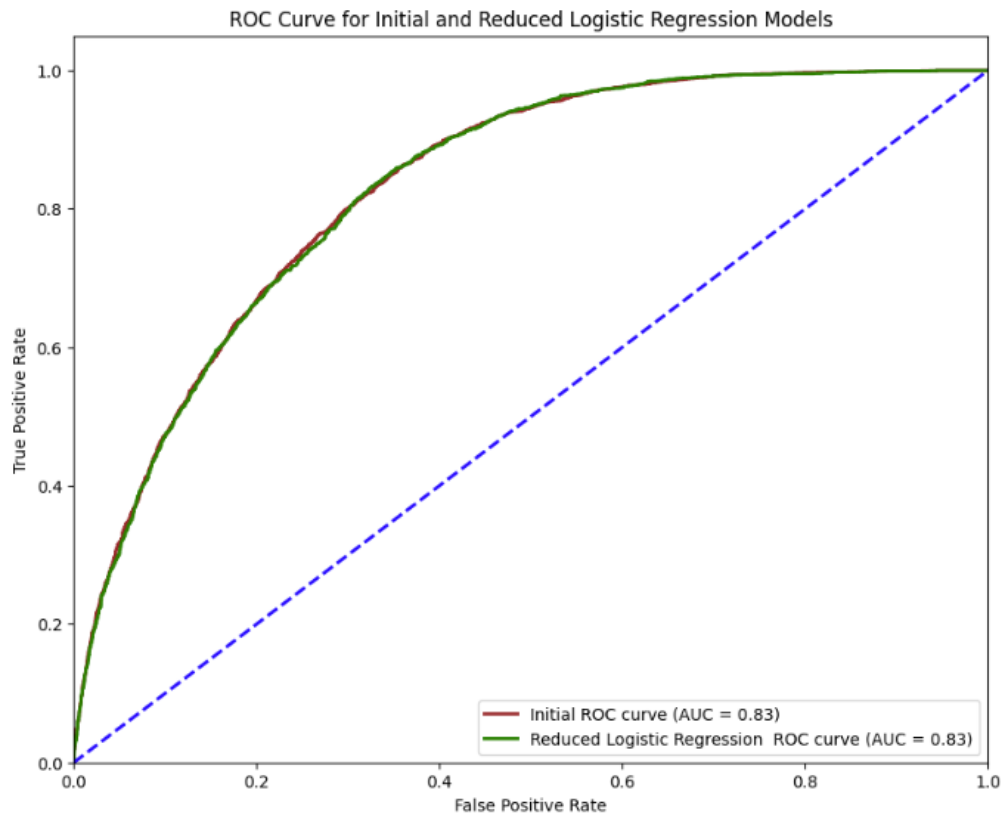
- reduced logistic regression model

Reduced Logistic Regression Model AUC: 0.8315380824027725

The AUC values for the initial model are higher than those for the Reduced model. However, the difference is negligible, and the two models perform well.

5. ROC curve

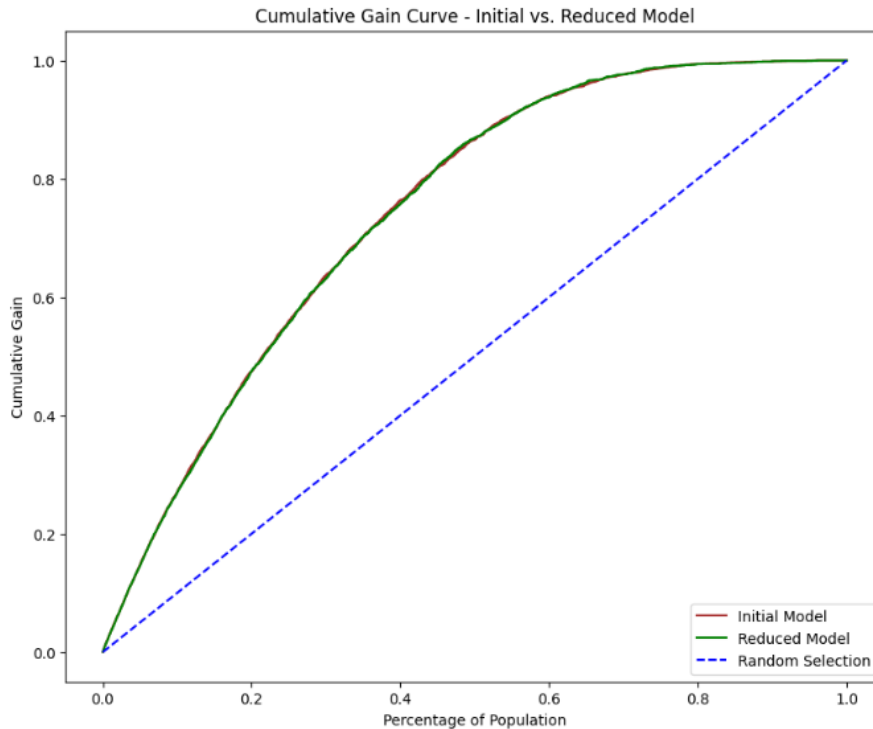
According to Bobbitt, Zach (2021, August 9), an ROC curve is a plot that displays the sensitivity and specificity of a logistic regression model.



The ROC curve plot comparing the performance of the Initial and reduced logistic regression models shows that both models have the same Area Under the Curve (AUC) of 0.83, indicating that their performance in distinguishing between positive and negative classes is equivalent. The curves for both models overlap almost entirely, showing no meaningful difference between the two in terms of classification ability. The blue dashed line represents random classification, and both models perform significantly better than random guessing.

6. Cumulative gains curve

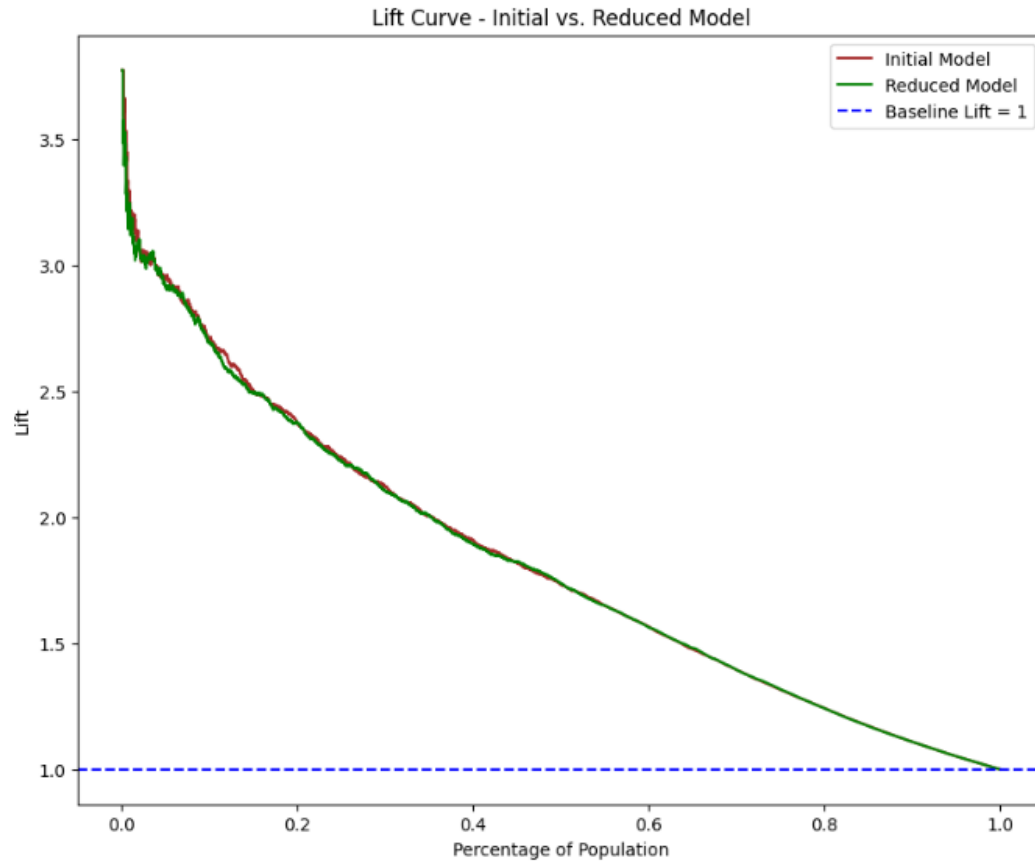
A cumulative gains Curve compares the performance of the initial model and the reduced models' performance against a random selection.



The initial and reduced models overlap almost entirely, suggesting that the feature reduction process effectively removes redundant or insignificant predictors without degrading the model's performance.

7. Lift curve

A lift curve compares the performance of the initial model and the reduced model in terms of their ability to "lift" or prioritize positive cases compared to random selection.



Both models demonstrate similar lift curves. The reduced model achieves comparable performance with fewer features, making it more efficient while maintaining the same ability to rank positive cases.

E2: OUTPUT AND CALCULATIONS

- confusion matrix

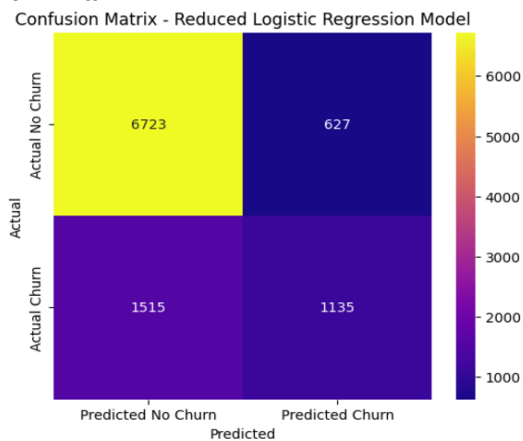
```
# Final Model Confusion Matrix
print("Reduced Logistic Regression Model Confusion Matrix:")
print(confusion_matrix(y, y_pred_final))

# Predict class labels (binary 0 or 1) for the final model
y_pred_final = final_result.predict(final_X) > 0.5 # Threshold the probabilities at 0.5

# Calculate confusion matrix
cm_final = confusion_matrix(y, y_pred_final)

# Plot confusion matrix using seaborn heatmap
plt.figure(figsize=(6, 5))
sns.heatmap(cm_final, annot=True, fmt="d", cmap="plasma", xticklabels=["Predicted No Churn", "Predicted Churn"], yticklabels=["Actual No Churn", "Actual Churn"])
plt.title("Confusion Matrix - Reduced Logistic Regression Model")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

Reduced Logistic Regression Model Confusion Matrix:
[[6723  627]
 [1515 1135]]
```



- accuracy calculation

```
# Predict class labels (binary 0 or 1) for the final model
y_pred_final = final_result.predict(final_X) > 0.5 # Threshold the probabilities at 0.5

# Final Model Accuracy
accuracy_final = accuracy_score(y, y_pred_final)
print(f"Reduced Logistic Regression Model Accuracy: {accuracy_final}")
```

Reduced Logistic Regression Model Accuracy: 0.7858

```
# Final Model Classification Report
print("Reduced Logistic Regression Model Classification Report:")
print(classification_report(y, y_pred_final))
```

Reduced Logistic Regression Model Classification Report:

	precision	recall	f1-score	support
0	0.82	0.91	0.86	7350
1	0.64	0.43	0.51	2650
accuracy			0.79	10000
macro avg	0.73	0.67	0.69	10000
weighted avg	0.77	0.79	0.77	10000

E3: CODE

Please modify the input and output file path parameters in the code

InputFilePath = 'C:/Users/beatr/OneDrive/Desktop/D208/D208_Data/Task2/churn_clean.csv'

Outputfile='C:/Users/beatr/OneDrive/Desktop/D208/D208_Data/Task2/Churn_df_encoded_Logistics.csv'



BeatriceMwangi_D2
08_DataModelling_1

F1: RESULTS

- a regression equation for the reduced model

$$\begin{aligned} \text{Log}((P(\text{Churn}=\text{Yes}) / (1-P(\text{Churn}=\text{Yes})) = & -3.7988 + 0.0128 \times \text{MonthlyCharge} + \\ & 0.1445 \times \text{Marital_Widowed} + 0.1599 \times \text{Gender_Male} + 0.5198 \times \text{Techie_Yes} - 1.6996 \\ & \times \text{Contract_One year} - 1.8450 \times \text{Contract_Two Year} - 0.8548 \times \text{InternetService_Fiber Optic} \\ & - 0.4411 \times \text{InternetService_Other} + 0.4453 \times \text{Multiple_Yes} - 0.1380 \times \text{OnlineSecurity_Yes} \\ & + 0.1618 \times \text{DeviceProtection_Yes} + 0.8662 \times \text{StreamingTV_Yes} + 1.0359 \times \text{StreamingMovies_Yes} \\ & + 0.1929 \times \text{PaymentMethod_Electronic Check} \end{aligned}$$

- an interpretation of the coefficients of the reduced model

Each coefficient in the model represents the log odds of a customer churning for a one-unit increase in the corresponding predictor, holding other variables constant. Here's a breakdown of the key coefficients:

- MonthlyCharge (0.0128): For every \$1 increase in the monthly charge, the log odds of churn increase by 0.0128, indicating a positive relationship with churn. This suggests that higher monthly charges are associated with a higher likelihood of churn.
- Marital_Widowed (0.1445): Widowed increases the log odds of churn by 0.1445, meaning that widowed customers are likelier to churn than others.
- Gender_Male (0.1599): Males have a higher likelihood of churning (coefficient: 0.1599), indicating a slight increase in the odds of churn compared to females.
- Techie_Yes (0.5198): customers marked as Techie_Yes are more likely to churn. The positive coefficient suggests that they are more likely to Churn.
- Contract_One year (-1.6996) has negative coefficients indicating that customers on one year are less likely to churn.
- Contract_Two Year (-1.8450) has negative coefficients indicating that customers on two-year contracts are less likely to churn.
- InternetService_Fiber Optic (-0.8548): Customers with fiber optic internet service are less likely to churn.
- InternetService_Other (-0.4411): Customers with Internet service others are less likely to churn.
- Multiple_Yes (0.4453): Customers with multiple services are likely to Churn.
- StreamingTV_Yes (0.8662) Customers with Streaming TV are likely to churn.
- StreamingMovies_Yes (1.0359) Customers with Streaming Movies are likely to churn.
- PaymentMethod_Electronic Check (0.1929): Customers paying via electronic check are slightly more likely to churn than other payment methods.
- OnlineSecurity_Yes (-0.1380) Offering online security reduces the likelihood of churn.

- DeviceProtection_Yes (0.1618) offering device protection increases the likelihood of Churning.

• the statistical and practical significance of the reduced model

Reduced Logistic Regression Model Summary:						
Logit Regression Results						
Dep. Variable:	Churn_Yes	No. Observations:	10000			
Model:	Logit	Df Residuals:	9985			
Method:	MLE	Df Model:	14			
Date:	Thu, 21 Nov 2024	Pseudo R-squ.:	0.2519			
Time:	23:20:52	Log-Likelihood:	-4325.8			
converged:	True	LL-Null:	-5782.2			
Covariance Type:	nonrobust	LLR p-value:	0.000			
	coef	std err	z	P> z	[0.025	0.975]
MonthlyCharge	0.0128	0.002	7.209	0.000	0.009	0.016
Marital_Widowed	0.1445	0.066	2.188	0.029	0.015	0.274
Gender_Male	0.1599	0.054	2.982	0.003	0.055	0.265
Techie_Yes	0.5198	0.069	7.566	0.000	0.385	0.654
Contract_One year	-1.6996	0.079	-21.650	0.000	-1.854	-1.546
Contract_Two Year	-1.8450	0.077	-23.928	0.000	-1.996	-1.694
InternetService_Fiber Optic	-0.8548	0.070	-12.175	0.000	-0.992	-0.717
InternetService_Other	-0.4411	0.077	-5.694	0.000	-0.593	-0.289
Multiple_Yes	0.4453	0.078	5.739	0.000	0.293	0.597
OnlineSecurity_Yes	-0.1380	0.056	-2.449	0.014	-0.249	-0.028
DeviceProtection_Yes	0.1618	0.058	2.785	0.005	0.048	0.276
StreamingTV_Yes	0.8662	0.099	8.749	0.000	0.672	1.060
StreamingMovies_Yes	1.0359	0.113	9.179	0.000	0.815	1.257
PaymentMethod_Electronic Check	0.1929	0.056	3.433	0.001	0.083	0.303
const	-3.7988	0.191	-19.917	0.000	-4.173	-3.425

The p-values for all coefficients in the reduced model are very small (typically below 0.05), indicating that these predictors are statistically significant in determining customer churn. The Pseudo R-squared value of 0.2519 indicates that the model explains around 25% of the variation in churn, which is reasonable for a logistic regression model. The Log-Likelihood Ratio (LLR) p-value of 0.000 confirms that the model is significantly better than a null model. The reduced model is more streamlined, focusing on significant predictors since the initial model includes many less relevant variables. The model is statistically significant, but the reduced model offers

better interpretability and efficiency by excluding variables with high p-values. Regarding practical significance, the reduced model highlights key factors that directly impact churn, making it more actionable for Churn.

- **the limitations of the data analysis**

Several limitations were encountered when analyzing this data. The analysis was limited due to a lack of understanding of the business data, how the data was collected, and the domain knowledge. More so, the data lacked appropriate dates to explore the effect of various features over a time period.

F2: RECOMMENDATIONS

The reduced logistic regression model identifies the following features MonthlyCharge, Marital_Widowed, Gender_Male, Techie_Yes , Contract_One year, Contract_Two Year, InternetService_Fiber Optic, InternetService_Other, Multiple_Yes, OnlineSecurity_Yes, DeviceProtection_Yes, StreamingTV_Yes, StreamingMovies_Yes, PaymentMethod_Electronic Check as the most important in Impacting the Churn.

Based on the results obtained from the reduced logistic regression model. The telecommunication company can predict the customers who will Churn 78.6 percent of the time. The logistic regression model output provides strong evidence that the model is a good fit for predicting Churn; however, more analysis is needed to improve model assumptions and residual behavior.

G: PANOPTO DEMONSTRATION

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=ac1597bf-f7bf-4fe0-8ca2-b23000713c7c>

H: SOURCES FOR THIRD-PARTY CODE

<https://stackoverflow.com/questions/11707586/how-do-i-expand-the-output-display-to-see-more-columns-of-a-pandas-dataframe>

I: SOURCES

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J: PROFESSIONAL COMMUNICATION